

The LLM Landscape for LMICs:

Overview, Evaluation Framework, Evaluation Components, Initial Recommendations

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Jigar Doshi*, Kashyap Jois, Akhil Kumar, Keith Hanna**

IPRD Solutions

and

P. Anandan**

AI Matters for Development

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* Consultant to IPRD Solutions

** Contact for correspondence: khanna@iprdgroup.com anandanpdot@gmail.com

EXECUTIVE SUMMARY

A new technology called “Large Language Models” has recently emerged from within the field of Artificial Intelligence that has indicated that machines can demonstrate human-like capabilities especially in areas such as question-answering, information retrieval, summarization and are able to conduct conversations with humans that resemble human-to-human conversations. LLMs have already begun to show their impact on many professions and walks of life. This document explores how LLMs can help enhance the lives of poor and underserved communities in Low-and-Middle-Income countries (LMICs). We are interested in understanding if and how LLMs can fill critical gaps in frontline support in primary health, agriculture and other domains that are due to insufficient numbers of trained experts to provide support and care to vast populations.

We show how to navigate the LLM landscape by demystifying them and explain how they are trained. We discuss how LLMs are being evaluated in their general use and we perform a survey of LLM models and tools. We identify the key challenges in using LLM-based capabilities in an LMIC context and suggest how these challenges can be overcome. We propose an approach for evaluating LLMs for LMIC applications and derive metrics that are specific to LLM usage scenarios and constraints. We illustrate the use of LLMs in LMICs through three example use-cases which are currently being actively pursued.

Success of LLMs in LMICs will depend not just on their capabilities but on how well they are tailored to the local context of usage (localization) and their deployment. We conclude with a set of recommendations for a course of action to safely bring the benefit of the power of LLMs to LMICs.

Localization:

- Incorporate local and contextual information about prevalent medical and environmental factors that can help understand the causes of the problems, the social, cultural, and economic factors that will affect what solutions may be feasible, and relevant traditional methods and best practices. This requires systematic documentation of such knowledge, and curation of the form of documents which are used during LLM fine-tuning.
- Ensure that the LLM solutions can communicate with the user and end beneficiary in the preferred language and dialect, in oral form, and are able to understand vernacular input and allow mixed mode communication including voice, text and offline voice notes.

Deployment:

- Adapt the LLM solutions to manage infrastructural issues such as connectivity, storage, limitations of digital system skills, and cost of deployment and maintenance.
- Incorporate change management programs to enable smooth transition from old processes and new processes.

- Ensure scalability and sustainability using local SW developers in LMICs for configuration, and SW developers worldwide to develop best-of-breed, core, reusable LLM software tools that solve specific issues that occur in LMICs.

Evaluation:

- Ensure that LLM metrics are developed for specific and relevant use cases in LMICs.
- Develop metrics to measure their success at guaranteeing Diversity, Equity, and Inclusion.
- Develop an approach to evaluating LLMs against systemic constraints on roles and authorities of various actors ranging from Front Line Workers to senior officials.

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1. Landscape Background, Objectives and Summary

1.1. Background

The primary purpose of this landscape is to obtain insight into the rapidly emerging breakthrough technologies in Artificial Intelligence (AI) called Large Language Models (LLMs) and to develop a framework to assess their use and application in Low and Middle-Income Countries (LMICs).

Within the last 15 years, the field of Artificial Intelligence has seen two remarkable breakthroughs that have already begun to show a large scale of social and economic impact globally across all aspects of life. The first was the potential of Deep Learning[1] (currently more commonly referred to simply as "machine learning"). Although the core principles of Deep Learning (ML) were discovered decades ago[2,3,4], it was not until recently that this technology has been shown to solve problems at scale in image recognition, spoken and textual language understanding, as well in a wide variety of domains such as education, commerce, economics, as well as in all areas of scientific research and R&D. However, it is the more recent innovation commonly known as "Large Language Models" (LLMs)[5,7] that has led to the widespread belief that we have reached the threshold where machines can demonstrate intelligent behaviors similar to that of humans and that the "era of Artificial Intelligence[54]" is here. As Bill Gates has recently said[6] "The Age of AI has begun".

Today, virtually every technological application and sector[55] is influenced by Machine Learning and Artificial Intelligence[15]. Commercial transactions are mediated by AI through recommendation algorithms[16] and fraud detection systems[10]. In news, media, and communication[19], AI algorithms curate personalized feeds and automate news summaries. Email prioritization is also AI-mediated, filtering spam and highlighting important messages[14].

In healthcare, AI facilitates diagnostic processes via image recognition technologies[24]. Banking employs AI for credit risk assessment[26] and fraud detection[10]. Educational platforms use adaptive learning algorithms to customize coursework[56]. In agriculture, drones equipped with AI surveil crop health, while manufacturing utilizes AI-driven robots for assembly line efficiency. The travel industry relies on AI for optimizing flight[27] and hotel bookings, and entertainment platforms like Netflix use AI algorithms for content recommendation[23].

Historically these AI models have focused on perception and understanding. More recently, however, Large Language Models have demonstrated the capability to generate human-like content[11,17]. The primary focus has been on linguistic and textual content, although there is rapidly growing research into the generation of similar quality spoken and image content. LLMs can read text and understand the information, write documents that are suited for a wide range of uses on almost any topic[12], write computer programs, generate visual and musical forms of artistic content[57], and create in a credible fashion. They are being used to augment human creativity and improve productivity across industries to solve the world's toughest problems.

As AI has already been transforming society and LLMs have already begun to do so, the vast potential and power of this technology to enhance the lives of citizens of Low- and Middle-Income Countries (LMICs) has already been recognized. For example, India's National Strategy for AI[28] has "AI for All", i.e., the inclusive growth of all sectors of the society through AI, as its goal. Likewise in a recent article, Mark Suzman, the CEO of the Bill and Melinda Gates Foundation (BMGF) notes that "Artificial Intelligence could drive progress for people" but notes that "… we must center equity and access" and proposes a set of first principles guiding the work of BMGF in AI. These principles include adherence to the core values of BMGF; promotion of co-design and inclusivity, proceeding responsibly, addressing privacy and security, building for equitable access, and ensuring transparency[29].

One major challenge in advancing the United Nations' Strategic Development Goals (UN SDGs) in LMICs is the shortage of trained experts, forcing reliance on under-equipped front-line workers (FLWs)[32,30]. Despite their dedication and customer trust, these FLWs often operate remotely and lack specialized support, compromising the quality and timeliness of decision-making and care[31,33,34,35].

LLMs have already demonstrated powerful linguistic and conversational capabilities. Among their core capabilities a few are particularly relevant for LMIC applications: conducting human-like conversation (Conversational capability), answering questions with accurate and relevant answers (Question Answering, Q-A), retrieving relevant information from a vast corpus of sources (Information Retrieval, IR), summarizing relevant information in a succinct manner (Summarization), language to language translation, sentiment analysis, and document writing.

1.2. Objectives

Given the power and potential of LLMs and the excitement around them, it is natural to explore whether this powerful rapidly evolving technology can help address the challenges faced in LMICs for providing advisory, triaging, and helping with critical care. In this document, we discuss what it would take to extend the power of LLMs to benefit lives in LMICs. Specific challenges and considerations uniquely related to LMIC applications include systemic constraints (such as connectivity), localized contextual data, the risk of providing misinformation and misguiding the users, culturally diverse users' needs, as well as equity and access. While these factors are certainly present in more affluent settings, they reach heightened levels of severity and complexity in LMICs. Also, the major development and investment in LLM technology is primarily anchored in High-Income Countries. This imbalance amplifies challenges related to both diversity and the naturally narrower focus of initial technological development. The situation underscores the necessity for a concerted effort to tailor LLM capabilities to meet the unique demands of LMICs for responsible and effective deployment.

In parallel with the rapid development of LLM technologies and the growing ecosystem around their development and usage, methods for evaluating the performance and utility of LLMs are also being developed. While these current evaluation methods are excellent for understanding the performance and limitations of LLMs used in general settings, LMIC applications have specific considerations that may not be adequately addressed by them. It is also the aim of this document to develop a method for evaluating Large Language

Models (LLMs) in Low- and Middle-Income Countries (LMICs). To achieve this, we initially survey existing evaluation frameworks for LLMs, select a particularly comprehensive one, and then suggest modifications to tailor it for LMIC scenarios.

Due to the lack of robust digital infrastructure in many LMICs, developing, deploying, and using LLMs in LMICs have to deal with challenges that go well beyond the capabilities of LLMs themselves. In this document we make recommendations for implementing digital LLM technologies in LMICs. These recommendations cover aspects such as solution integrity, robustness, scalability, adoption, and sustainability, all essential to long-term sustained impact. To illustrate, we cite specific real-world challenges and solutions from other digital initiatives in LMICs.

Lastly, we drill down further into the use of LLMs in LMICs, from the viewpoint of technology, evaluation, and implementation on the ground. An important aspect of our approach to studying use-cases of LLMs in LMICs is not to just provide an abstract, but to illustrate with three concrete examples.

1.3. Document Organization

The document consists of 4 major parts:

- Part I includes sections 2 to 5 and focuses on the LLM landscape at large.
- Part II includes sections 6 to 8 and dives into the use of LLMs in LMICs and discusses how to evaluate them for LMIC applications.
- Part III illustrates the use of LLMs in LMICs via three examples which are actively being pursued by different organizations. We use a common descriptive structure to define 3 different example use cases for LLMs in different sectors (Health and Agriculture). We choose a set of LLMs in terms of components and/or tools for different use cases with respect to various LMIC evaluation criteria, with one purpose being to show by example how the evaluation criteria can be used to pick LLM capabilities for particular use cases.
- Part IV summarizes the document and makes a set of recommendations for LLM applications in LMICs.

In section 2 we show how to navigate the LLM landscape, by demystifying “black box” capabilities. We achieve this by dividing them into components and describing how the components work together to make a functional LLM solution.

In Section 3 we explain how the LLMs are trained to achieve their impressive functional capabilities. We discuss the different ways in which LLMs can be interfaced by end users and system developers.

In section 4 we describe the challenges of implementing such an evaluation framework in the LMIC context and discuss one of the leading evaluation approaches for LLMs that are currently in practice.

In section 5 we describe how such an evaluation approach can be implemented, which includes defining and mapping LMIC-focused metrics onto other metrics that generic evaluation methods typically use.

In section 6 we answer the question why we believe LLMs are useful in addressing challenges in LMICs.

In section 7 we discuss what may be required in order to build LLMs suited for LMIC applications, what it would take to deploy, support and scale them in LMIC environments, and what it would take to make the user experience suitable and engaging for the LMIC user. We also identify the key issues of using LLM-based capabilities in an LMIC context and suggest how putting programmatic development effort into different LLM component areas can potentially make an impact. One of our objectives is to demystify LLMs and their components so that as new LLM capabilities are proposed or announced, then their impact (or not) on the LMIC LLM ecosystem can be understood.

In section 8, we discuss evaluation criteria for each of the topics covered in section 7, namely building LLMs with the right set of capabilities, deploying and scaling them and making them usable for the LMIC audience.

In addition, we have three Appendices.

In Appendix A, we discuss the learning mechanisms of LLMs. In Appendix B we discuss a possible set of hierarchical requirements and an evaluation framework with the goal of maintaining some consistency and re-use of requirements and evaluation criteria across different use-cases in different geographies. High-level organizational requirements and evaluation criteria are at the top of the hierarchy, and specific use-case requirements and evaluation criteria are at the bottom. Finally, Appendix C includes an elaboration of the recommendations that were made in Part IV and proposes next steps for their implementation.

PART I - NAVIGATING THE LLM LANDSCAPE

2. Introduction to Large Language Models (LLMs)

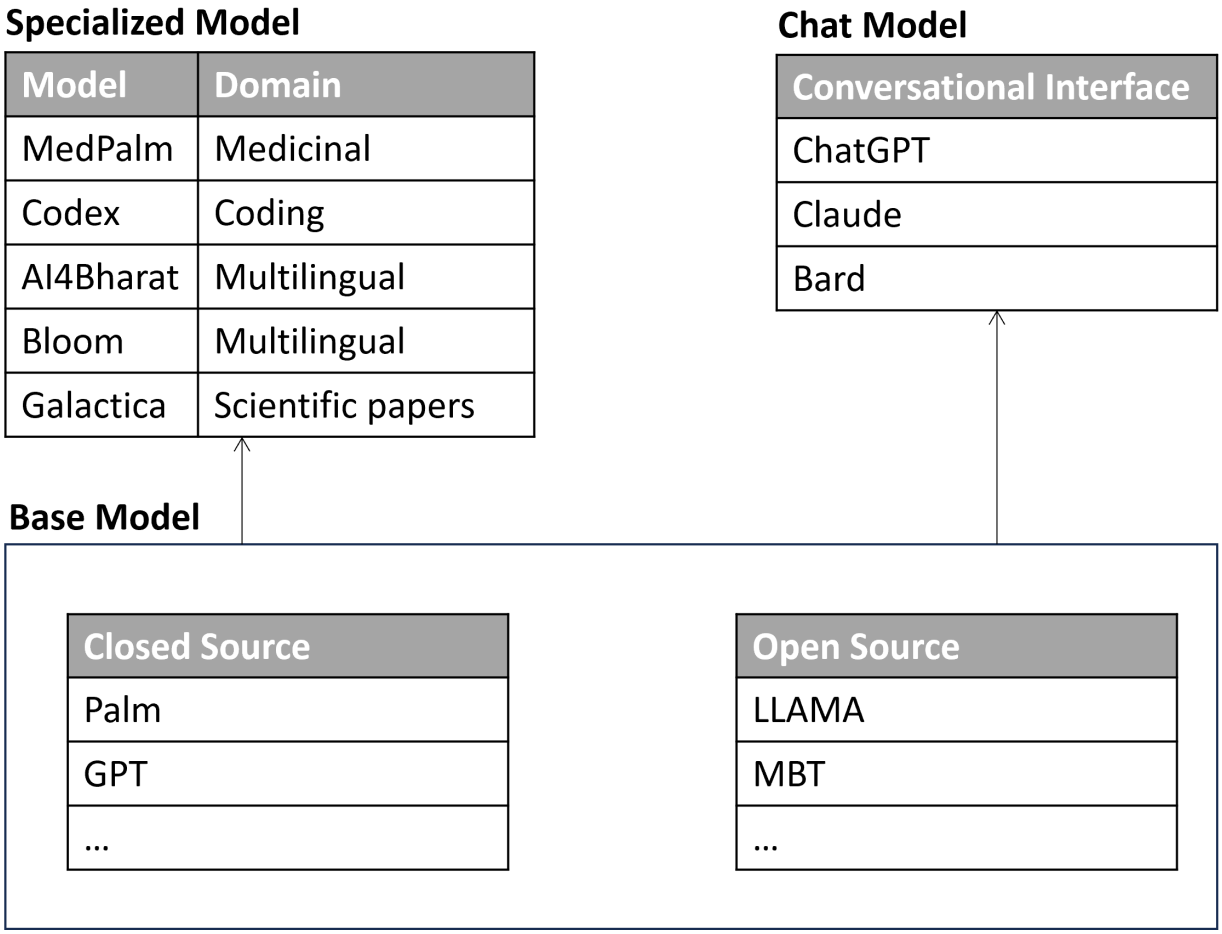
Large Language Models (LLMs) are a significant advancement in the field of Artificial Intelligence (AI), specifically in Natural Language Processing (NLP). These models exhibit unprecedented text generation and understanding ability; making them capable of performing a wide range of tasks that require an understanding of language.

Examples of LLMs include GPT[11,17,58,59] (Generative Pretrained Transformer) by OpenAI, PALM (Pre-training with Autoregressive Language Modeling)[60] by *Google* and LLAMA (Large Language Model)[61] by *Meta* and Falcon by TII[70,71].These models have been trained on vast amounts of text data, enabling them to generate coherent and contextually relevant responses in a wide class of usage scenarios.

LLMs are typically situated within the context of an application. They serve as the backbone of systems that can perform various tasks such as answering questions, writing emails, creating poems, and extracting information from documents. The versatility of LLMs stems from their ability to understand and generate text, making them a powerful tool for a multitude of applications.

In the context of Low and Middle-Income Countries (LMICs), LLMs hold significant potential. They can be used to provide information and services in local languages, support education, literacy initiatives, and even aid in administrative tasks. However, the effective deployment of LLMs in these contexts requires a deep understanding of their capabilities, and limitations, and tailoring them to the specific needs and constraints of LMICs.

As illustrated in the figure below, LLMs consist of “base models” which can either be open-source or closed-source models. These base models are generic and have been trained on data at the internet scale and can be applied to a wide range of tasks across many domains. More recently, however, “specialized” LLMs have been developed that provide a deeper level of information on specific domains such as medicine and computer programming. All these models are typically accessed by end users via textual conversation interfaces, labeled “Chat Model” below.



3. Training Process and Interfaces for LLMs

The training process of Large Language Models (LLMs) typically involves two crucial stages: base model training and fine-tuning.

3.1. Base Model / Unsupervised Learning

The bedrock of LLM development is established during the base model training stage, an intensive process often described as unsupervised pre-training. In this phase, iconic base models like GPT, PALM, LLAMA, Falcon, and their subsequent iterations are trained on colossal datasets that are primarily sourced from public internet repositories and private databases. The exact constituents of these datasets remain largely undisclosed, particularly for closed-source models.

The training employs a form of unsupervised learning that trains these models on trillions of sentences. Here, the models are tasked with predicting subsequent words in sentences, a method through which they learn patterns, structures, and intricate relationships within the data. Through this simplistic next word prediction of trillions of words during training emerges surprising abilities in the model to comprehend and

generate text across a diverse array of topics. The approach offers a stark contrast to fine-tuning, a subsequent stage that typically requires time-consuming and costly human labeling of the desired outputs to train the model.

It's worth noting that the true power of LLMs manifests with scale[39,62]. Even as the underlying training recipes remain similar, it's the act of scaling that amplifies the model's capabilities, allowing for emergent properties[40] and skills that are not present in smaller, less sophisticated models. This initial training process is both computationally intensive and data hungry, often requiring state-of-the-art GPU clusters running for weeks or even months. Notable examples include the recently released LLAMA-2 model from Meta, which was trained on 1.4 trillion tokens over 21 days, utilizing 2048 A100 GPUs at an estimated cost of around $5 million USD according to commercially available cloud rates.

Models developed through this unsupervised training, such as the GPT series by OpenAI, have shown that emergent skills[40] are unlocked when scaling compute and data. Open-source models like LLAMA and Falcon have also contributed to the landscape, serving as platforms for further research and customization. Financially, the investment in this foundational training is substantial, often running into the millions. However, the yield is a versatile and adaptable base model that is not only capable of understanding and generating complex text but also ripe for further fine-tuning and specialization in later stages.

3.2. Fine Tuning Model / Specialization

Fine-tuning is akin to the postgraduate education for Large Language Models (LLMs), where a generalized scholar—adept but a jack-of-all-trades—transforms into a subject matter expert. It's a deliberate, meticulous process that refines the broad capabilities of a base model into a specialized tool with a focused skill set.

The objectives of fine-tuning are twofold. First and foremost, it serves as a rigorous quality control phase. This is where we meticulously teach the model to avoid harmful, offensive, or otherwise unwanted output. Safety isn't just a feature; it's a mandate. In the context of conversational agents like ChatGPT, this training becomes particularly intensive. A two-step approach is applied: initial instruction-based fine-tuning, followed by RLHF, or Reinforcement Learning from Human Feedback. The former lays down the basic guidelines for the chatbot, teaching it the dos and don'ts of conversation. The latter fine-tunes these guidelines based on real-world interactions and human evaluations, making ChatGPT not just a safe conversationalist but a genuinely helpful, context-aware assistant.

The second objective is specialization. This is like turning a well-rounded athlete into a sprinter, high-jumper, or marathon runner. We might, for instance, train Google's PALM2 to become Med-PALM[63], an oracle for medical advice, or adapt it into AI4Bharat[36] to assist in multiple Indian languages. Specialization can range from targeting a specific industry, like healthcare, to honing skills for a specific task, such as programming assistance via models like CODEX[64].

However, the complexity of fine-tuning shouldn't be underestimated. It's a resource-heavy process that demands specialized datasets, substantial computational power, and a deep mastery of machine learning techniques. Access to the architecture of the model is

crucial—you can't fine-tune what you can't see or touch. Financial costs can escalate, especially if specialized human expertise, like medical consultants for Med-PALM, is required for accurate annotation.

Parameter-Efficient Fine-Tuning (PEFT)

As previously indicated, the conventional fine-tuning approach employed by organizations like OpenAI or Google is both computationally demanding and reliant on costly human annotations. This makes the method challenging to implement and often financially prohibitive, particularly in resource-constrained settings such as Low and Middle-Income Countries (LMICs). To address this challenge, a specialized subfield within machine learning focused on parameter-efficient fine-tuning has emerged in recent years.

Some of the most prominent methods in this subfield are emerging techniques like Low-Rank Adaptation[65] (LoRA) and P-tuning v2[66]. These have shown transformative potential in the realm of Parameter-Efficient Fine-Tuning (PEFT). Specifically, LoRA can reduce the number of trainable parameters to as low as 0.01% in large models like GPT-3 without sacrificing performance. Similarly, P-tuning v2 requires only between 0.1% to 3% of trainable parameters per task and still outperforms traditional fine-tuning methods across a range of model sizes.

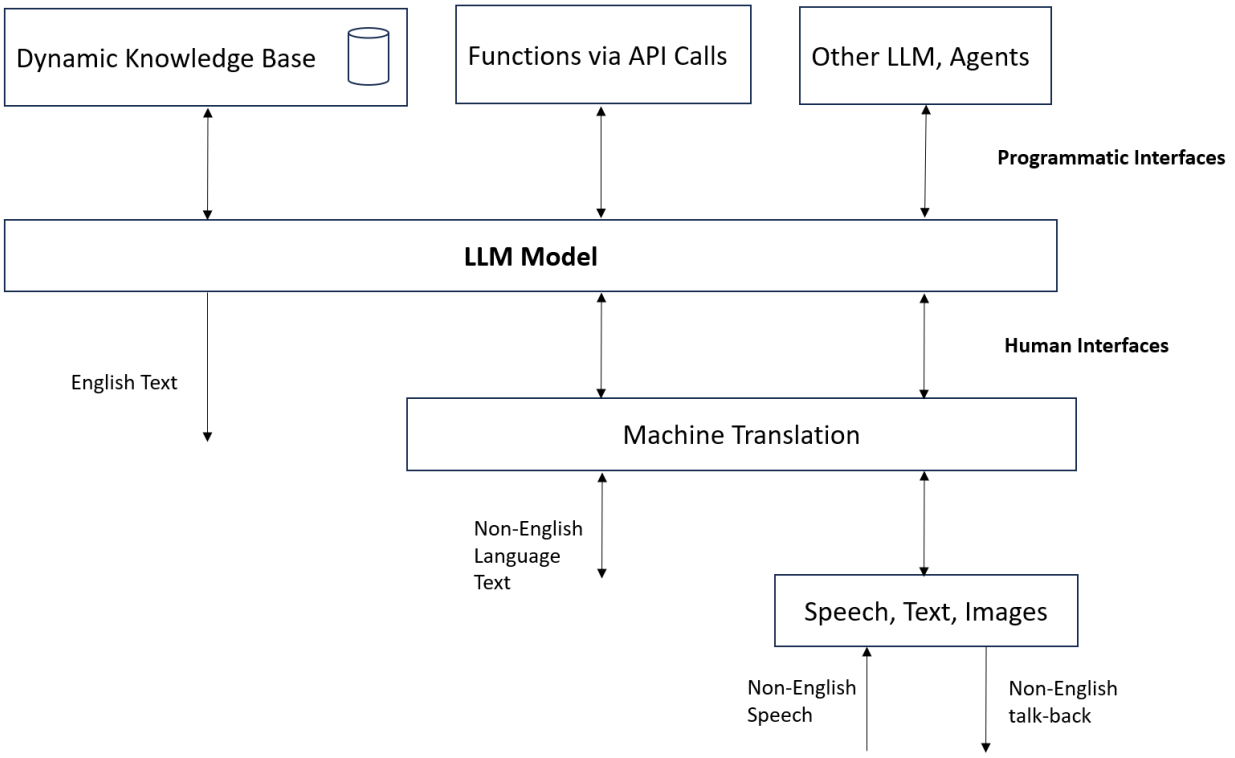
PEFT techniques play a crucial role in democratizing access to fine-tuned Large Language Models (LLMs), particularly in LMICs. By substantially reducing both computational and financial requirements, these methods enable the deployment of specialized applications that would otherwise be infeasible due to resource limitations.

In summary, the exhaustive base model training lays the versatile foundation, while meticulous fine-tuning sculpts specialized expertise. The former is computationally intensive but creates a jack-of-all-trades; the latter demands care and resources but births subject matter experts. Emerging techniques like parameter-efficient fine-tuning promise to democratize access to this transformative technology. However, crafting helpful, harmless AI still requires responsible partnerships between researchers, developers, and society. Training is but the first step; deployment and oversight determine whether AI's potential is purposefully realized.

	Base Model	Chat Model	Specialized Model
Data Used	Internet Scale. Trillions of words	Online conversation, Quora, Reddit	Small, curated corpus of specialized knowledge
Compute Power	100's millions of USD$ compute	10's thousand USD$ of compute	10's thousand USD$ of compute
Human Annotation	No Annotation	1000's of hours of annotation	1000's of hours of domain experts
Capabilities	Generic vocabulary and grammar as a vehicle to communicate questions and answers, as well as generic knowledge applicable both in LMICs and non-LMICs	Knowledge of human interaction etiquette, e.g. avoiding hateful speech	Specific technical information and knowledge

3.3. Interfaces to LLMs

Interfacing with Large Language Models (LLMs) can be done in various ways, depending on the specific needs of the user or the application. These interfaces can be broadly categorized into two types: human interfaces shown between the 2nd and 3rd rows of block elements in the diagram below, and programmatic interfaces shown between the 1st and 2nd rows.



Human Interfaces

Human interfaces are how users interact directly with the LLM. The simplest form of this is a text-based interface, where a user types a query or command, and the LLM responds with generated text. This is the primary mode of interaction for models like ChatGPT.

However, in contexts where literacy rates may be low or where typing is not the most convenient mode of interaction, other forms of user interfaces can be employed. For instance, an audio interface can be wrapped around the LLM, allowing users to interact with the model using speech. This can significantly improve accessibility and ease of use.

Another important aspect of user interfaces is language support. While LLMs are typically most proficient in English, their performance can decrease significantly for low-resource languages. To address this, specialized models like AI4Bharat[36] can be used to add multilingual functionality to the LLM. This allows users to interact with the model in their native language, further enhancing accessibility.

Programmatic Interfaces

Programmatic interfaces allow LLMs to be integrated with other systems or models. This can significantly enhance the capabilities of the LLM and allow it to be used in more complex applications.

One common form of programmatic interface is Retrieval Augmented Generation (RAG). This allows the LLM to be augmented with a “dynamic knowledge base” that contains custom and use-case specific data, bringing the model up to date and adding local contextual information. This can also help ensure the factual correctness of the model's outputs.

LLMs can also be chained with other models. For instance, a general-purpose model like ChatGPT could defer certain questions to a specialized medical LLM. This allows the system to leverage the strengths of multiple models, providing more accurate and relevant responses.

Finally, LLMs can be connected with other software systems. This can help overcome some of the limitations of LLMs, such as their difficulty with tasks like counting or sorting elements. By augmenting the LLM with traditional software functions, the system can provide more reliable and accurate outputs.

In conclusion, the interfaces to LLMs are a crucial aspect of their deployment and use. By carefully designing and implementing these interfaces, we can ensure that LLMs are accessible, useful, and effective in a wide range of contexts.

4. Evaluation Framework

4.1. Introduction

Evaluating Large Language Models (LLMs) is a complex and multifaceted process. It involves assessing the models' performance across a wide range of tasks, domains, and languages, and using a variety of metrics to capture different aspects of their performance. The goal is not just to determine which models perform best in a narrow, technical sense, but also to understand how they interact with the broader social, cultural, and ethical contexts in which they are deployed. This evaluation process is crucial for identifying the strengths and weaknesses of different models, guiding their development and deployment, and ensuring that they are used responsibly and effectively. In the following sections, we delve into the specifics of how LLMs are evaluated, focusing on the models themselves, the metrics used to assess them, and the scenarios or benchmarks that define their operational contexts.

The evaluation of LLMs often involves a comparison of different types of models. These can be categorized based on whether they are open-source or closed-source, and whether they are domain-specific or general-purpose. (*To recap from previous sections, open-source models are those whose code and architecture are publicly available, while closed-source models are proprietary, and their details are not publicly disclosed. Domain-specific models are designed to perform well on tasks within a specific domain or field, while general-purpose models are designed to handle a wide range of tasks across different domains.*) For a comprehensive evaluation, it would be beneficial to pick a mix of models, that include base models, chat models, and domain-specific models.

In this section we review the emerging evaluation approaches and frameworks that are being developed in the general AI domain. In Section 8, we will look at how to adopt and adapt such frameworks for LMIC uses and applications.

4.2. Challenges

Evaluating LLM-based capabilities, particularly those with broad applications such as text-to-text models, presents a unique set of challenges. These models are designed to generate a wide array of outputs, ranging from precise responses to more ambiguous, fuzzy outputs. This variability in output complicates the process of evaluation, as it becomes difficult to establish a single standard of correctness or usefulness.

Furthermore, when these models are applied to diverse tasks such as question answering, summarization, information retrieval, reasoning, planning, math, and code generation, the complexity of evaluation increases. Each of these tasks has inherent ambiguity and multiple correct answers, making it challenging to measure the model's performance accurately and consistently.

The complexity of assessing language models is compounded by limitations in human evaluation, often regarded as the gold standard. Such evaluations can be subjective and may fail to capture the full spectrum of a model's capabilities. Evaluators may diverge in their criteria for what constitutes an accurate or superior response, a divergence influenced by factors such as background, expertise, and personal biases. This underscores the necessity for more objective, robust evaluation methodologies capable of addressing these issues.

4.3. Evaluation frameworks and approaches

The domain of Artificial Intelligence has a long tradition of evaluating the performance of various algorithms, models, and systems through the use of benchmark datasets. Peer review mechanisms have ensured that AI algorithms and models require evaluations for them to be published and/or widely disseminated. As a result, it is in the interest of the creators of the models to also collect (and publish) benchmark datasets and show the performance of their models against these benchmarks. This tradition has continued with the Large Language Models. However, these benchmarks are definite by specific choices of scenarios and metrics that vary from model to model[48,49,50]. Different benchmarks make different decisions on what to prioritize, how to make these decisions, and to what extent these processes are made clear in presenting the benchmark. Even the metrics chosen typically vary across the models and evaluations. Accuracy is perhaps one metric that is most often considered by a variety of LLM evaluations. However, there are other metrics that are important. For instance, a model may excel at producing answers which may be accurate in a limited context or suitable for a particular audience segment (e.g., men, a particular ethnic group, a particular set of nationalities, etc.), but may be incorrect or even considered biased or toxic when applied to other audience segments. Hence, a multi-pronged assessment that captures the full spectrum of a model's capabilities and limitations is imperative.

The recent HELM (Holistic Evaluation of Language Models) framework paper[22], from Stanford University represents a focused and comprehensive examination of Large Language Models (LLMs). This framework distinguishes itself through its cohesive set of evaluation criteria applied to a wide range of models. The HELM approach rigorously specifies datasets, use-cases, tasks, models, and performance metrics, thereby establishing a standardized protocol for in-depth analysis. Such standardization facilitates nuanced model comparisons and identifies opportunities for improvement.

HELM delineates a detailed classification of "scenarios" and performance metrics, thereby providing a clear operational framework for LLM applications. For example, a specific "scenario" could be the utilization of an LLM in answering medical queries in English.

In its assessments, HELM employs seven essential metrics over 16 fundamental scenarios, along with specialized evaluations for particular facets. The research assessed

30 notable LLMs across 42 scenarios, thereby establishing an extensive performance benchmark for the field.

Seven Metrics Definitions:

Accuracy is the primary metric that most researchers focus on when evaluating Large Language Models (LLMs). It broadly measures how well the model understands and processes language across various tasks. For example, in text classification, accuracy might be quantified using the F1 score, a balanced measure that considers both precision and recall. This metric is central to understanding the model's ability to correctly classify texts, making it a key aspect of LLM evaluation.

Calibration, or the model's ability to express its uncertainty, is another important metric. In high-stakes settings where models inform decision-making, it's crucial that they can anticipate and accommodate their errors. This allows users to understand the confidence level of the model's predictions and make informed decisions accordingly.

Robustness is a measure of how stable a model's predictions are under small, semantics-preserving perturbations, and how semantics-altering perturbations influence model behavior. A robust model should be able to handle variations in the input without significant changes in its output.

Fairness and *bias* are two closely related metrics. Fairness refers to disparities in task-specific accuracy across social groups, while bias refers to systematic asymmetries in language choice. A fair and unbiased model should treat all social groups equally and avoid perpetuating harmful stereotypes.

Toxicity is an instance-level property of text, encompassing related concepts like hate speech, violent speech, and abusive language. A model that generates toxic outputs can cause harm and discomfort to users, even if it's accurate in a technical sense.

Efficiency is a measure of how well a model uses computational resources during both training and inference. Expensive training and inference costs can make models less usable and less accessible to a wide range of users. Therefore, efficient models are more desirable from a practical standpoint.

Scenarios, Datasets and Tasks

In the HELM framework[22], a "scenario" (as depicted in the second column of the table below) is a multi-faceted concept. It's shaped by the specific task at hand (first column in the table), the domain of application, and the language in use. The task outlines the system's primary function — be it question-answering or summarization — while the domain specifies the data landscape, such as Wikipedia or clinical notes. The language, be it English or Arabic, sets the linguistic stage. Scenarios are, in essence, the real-world arenas where we envision the language model will deliver capability. They serve as the scaffolding for evaluation, chosen based on criteria like coverage, minimality, and user-centricity. Picture, for instance, a scenario where a language model tackles question-answering in English, drawing from contemporary clinical notes. It should be noted that in the realm of language model evaluation, the term "scenario," as well as other terminologies, are specifically coined in the HELM paper and are not yet standardized

within the broader Natural Language Processing (NLP) community. However, we find this to be a useful framework to structure evaluations and potentially extensible, hence we propose using it for our purposes.

Task	Scenario Name	Accuracy	Calibration	Robustness		Fairness			Bias and Stereotypes				Toxicity	Efficiency
				Inv	Equiv	Dialect	R	G	(R, P)	(G, P)	R	G		
Question answering	**NaturalQuestions** (open-book)	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
	NaturalQuestions (closed-book)	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
	NarrativeQA	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
	QuAC	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
	BoolQ	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	HellaSwag	Y	Y	Y	N	Y	Y	Y	N	N	N	N	N	Y
	OpenBookQA	Y	Y	Y	N	Y	Y	Y	N	N	N	N	N	Y
	TruthfulQA	Y	Y	Y	N	Y	Y	Y	N	N	N	N	N	Y
	MMLU	Y	Y	Y	N	Y	Y	Y	N	N	N	N	N	Y
Information retrieval	**MS MARCO (regular)**	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
	MS MARCO (TREC)	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
Summarization	**CNN/DailyMail**	Y	N	N	N	N	N	N	Y	Y	Y	Y	Y	Y
	XSUM	Y	N	N	N	N	N	N	Y	Y	Y	Y	Y	Y
Sentiment analysis	**IMDB**	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Toxicity detection	**CivilComments**	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
Miscellaneous text classification	**RAFT**	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y

Scenarios-metrics matrix. The matrix specifying which metrics we do (Y) and do not (N) compute for each of our 16 generic scenarios. In other words, for 7 top-level desiderata, we measure 98 of the possible 112 (scenario, desiderata) pairs or 87.5%. For the remaining 14 pairs, the majority are not well-defined (e.g. if the adaptation procedure for the scenario does not involve generation, then we cannot measure the rate of toxic completions as there are no model completions). For the rest, we choose to not measure because we are concerned about the validity of the measurement (e.g. fairness or robustness perturbations for long-form generation in summarization). **Abbreviations:** Invariance, Equivariance, Race, Gender, Professions

To unpack this further, let's delve into the building blocks of a scenario: datasets and tasks. Datasets are curated collections of examples within a specific knowledge domain. Crafting a dataset can range from labor-intensive human annotation to straightforward web scraping for public data like Wikipedia articles.

Tasks, conversely, are the specific roles we assign to the language model. Take question-answering as an example: the model receives a query (input) and is expected to furnish a precise answer (reference). Imagine a healthcare worker leveraging a language model to tap into medical insights. A query like "What are the symptoms of diabetes?" would prompt the model to deliver a succinct, accurate response, such as "Frequent urination, increased thirst, and unexplained weight loss."

Scenarios are then scored, serving as performance indicators for the model in relation to the task, domain, and linguistic variety. High scores are accolades, while low scores are red flags.

The table above encapsulates this intricate web of scenarios and metrics, offering a panoramic view of Helm’s evaluation landscape.

However, two important caveats warrant immediate attention:

1. The LLM revolution is in its infancy, barely 2-3 years old. A universally accepted evaluation framework is still on the horizon. The methodologies discussed here are snapshots of the current state-of-the-art and should be viewed as provisional.

2. More importantly, our efforts necessitate a focus on target use cases and scenarios specifically relevant for LMIC applications, a domain often overlooked by mainstream LLM research. In the LMIC context, key values like equity, access, and inclusion take on unique meanings that can vary dramatically depending on the use case. Additionally, a one-to-one mapping between organizational definitions and those researched and developed by other groups is unlikely. Therefore, a specialized process must be developed to align organizational definitions with those from external research, enabling the leveraging of such work.

It's crucial to acknowledge that this evaluation framework, while robust, may not fully capture the real-world nuances of large language models. Their true utility often unfolds over extended interactions across multiple sessions.

Moreover, in the LMIC context, system and societal constraints add another layer of complexity to evaluations. These limitations highlight the need for a holistic, system-level assessment that contemplates the broader societal impact of these models.

5. A Survey of LLM Models and Tools

This section lists the LLMs models and software tools available at present. We divide these into three categories: LLM Base models, LLM Specialized models, and Software tools and infrastructure useful to programmatically interface with and use LLMs for a broad variety of applications. In the following sections we define requirements for these based on LMIC usage scenarios and use-cases and provide a preliminary assessment of the LLMs and tools based on these requirements and discuss ways to take a systematic approach to their evaluation.

5.1. LLM Base Models

The LLM Base models form the foundation for all other LLM models and incorporate vast general knowledge gathered from an extremely large corpus of documents and other sources of information available online. These models also show the emergent behaviors that have led to the belief that these models are artificially intelligent. The base models can be divided into closed and open-source models.

Closed: These are proprietary models developed by organizations and often serve as the foundation for a myriad of applications within the LLM landscape. Their access is restricted, which might limit customization options.

Examples: GPT-4, Claude 2, PaLM

Open Source/Free access: Open models democratize access to LLM technology, allowing for widespread research, innovation, and adaptation. They serve as a cornerstone for academic and community-driven advancements in the LLM landscape.

Examples: Llama2, Falcon, Bloom, Open Assistant, Mistral

5.2. LLM Specialized Models

The LLM Base models, while containing vast knowledge and being capable of emergent intelligent behavior, are not directly accessible to end-users. So, at the heels of the

development of these models there was an emergence of specialized models that facilitated interaction with and use of the base models. The most widely used models can be classified as “General Purpose Assistants” which provide direct user-level interfaces to LLMs. As in the case of LLM Base models the specialized models include both open and closed-source models.

General Purpose Assistant: These models are designed to simulate human-like conversations. Whether closed or open, they are integral to offering user-friendly interactions in the LLM ecosystem.

Examples (Closed): ChatGPT, Claude2, Bard, HeyPi, YouChat.

Examples (Open): Open Assistant, *BloomZ*

At the same time, it was recognized that models that specialize for particular subject domains or task domains might be useful for a wide variety of usages and applications. Such special skills models include:

Medical: Tailored for medical queries and information dissemination, these models are pivotal in healthcare applications within the LLM landscape.

Example: Med Palm

Coding: Designed to aid developers, these models can suggest, correct, or even write code. They're becoming indispensable in the software development workflows.

Examples (Closed): GitHub Co-pilot, Codex

Example (Open): Stable Code

Interaction in non-English languages: With a focus on linguistic diversity, these models cater to regional languages, ensuring the LLM landscape is inclusive and globally relevant.

Examples: AI4Bharat[47], M2M-100, Bloom 1000 language, Cohere model

Multimodal: Processing multiple types of input, these models are pushing the boundaries of the LLM landscape, merging text, image, and sometimes audio processing.

Examples: Flamingo, Gato, Fuyu

Robotics: Tailored for controlling or aiding robotic systems, these models are a testament to the versatility of the LLM landscape.

Example: RT2

As might be expected this list is continuously expanding as new models special to various domains and applications such as law, finance, and entertainment are emerging, underscoring the adaptability and vast potential of LLMs.

5.3. Software Tools and Infrastructure

Along with the development of new and improved base and specialized models as well as brand new models, a growing software ecosystem has been developing in order to enable developers to programmatically interact with these models and to make the process of deploying these models and developing new ones smooth. Below we describe some of these software ecosystem components that have emerged.

Core Software Tools: These are essential utilities that facilitate model training, data processing, and integration with private databases, ensuring the robustness and adaptability of the LLM ecosystem.

Examples: TensorFlow, PyTorch, Jax

Vector Databases (Embedding): These databases specialize in storing and retrieving vector embeddings, enabling efficient data access, and enhancing real-time LLM responses. These are particularly useful in working with private databases or databases that are specific to a particular domain.

Examples: Pinecone, ChromaDB, Weaviate, Milvus

API-Plugins: By allowing LLMs to connect with external services, plugins expand the functionality and integration capabilities of models, making them more versatile and user-centric.

Examples: Wolfram, Zapier, Weather.com

Infra: Compute/Cloud (Infra): These platforms provide the computational backbone of the LLM ecosystem, enabling scalable training and deployment of models.

Examples: AWS, GCP, Azure, Databricks, Mosaic, Anyscale

Data Pipelines: Tools that streamline data flow and processing, ensuring real-time responsiveness and efficient model training in the LLM landscape.

Examples: Databricks, Airflow, Kafka

Hub: Serving as repositories and collaboration platforms, hubs foster innovation and sharing in the LLM community.

Examples: LLM Resource Hub, Hugging Face

Orchestration: These tools handle the deployment, scaling, and management of LLM services, ensuring seamless operation and integration in diverse applications.

Examples: Langchain, Kubernetes

More recently we are also seeing the emergence of software tools that support the evaluation of LLMs:

Model Eval Tools:

LLM Logs: Platforms for tracking model training, evaluating performance metrics, and ensuring continuous improvement in the LLM ecosystem.

Examples: Weights and Bias, MLFlow

LLM Validation Tools: Tools that critically assess model outputs against desired standards, ensuring reliability and trustworthiness of LLMs in real-world applications.

Examples: Guardrails, Microsoft Guidance.

PART II - USE OF LLMS IN LMICs

6. Why LLMs?

We now turn to the potential ways in LLMs can be used to address deep and systemic challenges in LMICs, especially towards improving healthcare in poor and underserved communities, alleviating poverty by reducing loss and enhancing productivity in agriculture in small-hold farms which are typical in vast parts of LMICs and improving literacy and providing basic education to billions of people in poor communities.

One of the primary and continuing challenges in addressing many of the UN SDG goals towards health, poverty reduction and education is the lack of adequate human capacity to serve the needs of poor communities. For example, public health systems in LMICs typically have insufficient number of doctors and other trained medical experts and hence rely on a vast system of front-line or community health workers (FLW/CHW or ASHAs in India) who are often supervised by nurse-midwives, who are themselves often insufficient in number and in their capacity to provide care.

LLMs can be used in a variety of ways to address these gaps in primary health care in LMICS. For instance:

1. LLMs can assist CHWs in patient triaging, thereby enabling the patients who need further care to be sent to clinics and to reduce the number of false alarms. This can be for particular health challenges or disease conditions or ensuring safe pregnancy and delivery for mothers.
2. LLMs can be used to send timely and personalized nudges to the patients under care. This will help improve drug regimen adherence and reduce incidence of loss-to-follow-up.
3. LLMs can be useful in reducing the administrative burden placed on CHWs such as form-filling and reporting. This can ensure that care is provided to eligible patients as well as in the collection of vital data about the patient's health and medical condition.

In all these cases, LLMs can augment the capabilities of the CHWs, thereby significantly enhancing their capacity to provide care and to increase their effective reach.

Similarly, success in agriculture often depends on being able to address rapidly changing dynamic conditions in weather, soil health and other environmental factors. Losses due to pest infestation are often unpredictable due to varying causes and levels over seasons, and the need to balance the timely use of pesticide against the potential toxic effects of these chemicals. Traditionally, farmers receive support from public extension systems, which typically deliver advisory via field demonstrations, or in-person training facilitated by frontline workers (FLWs) also known as Extension Agents (EA). These practices are not cost-effective at scale, vary greatly in quality, and largely disseminate generalized top-down driven advisory, with major gaps in and lack of consideration for gendered extension messages. In addition, the system faces other challenges that include limited reach, language, and communication barriers, limited private sector involvement, and lack of evaluation mechanisms.

A natural solution is to use LLMs to build an AI-powered chatbot to understand the needs, preferences, and values of the farmers and increase their agency by engaging them as primary users while allowing Extension Agents to provide the proper interface to the EA system and the relevant government programs.

Examples of user-needs that can be addressed by LLMS in agriculture include:

1. Timely, contextualized, and customized advice to farmers on a variety of topics such as: soil management, crop rotation, adapting to weather and climate variations, fertilizer usage, pest infestation control, pesticide usage, etc.
2. Reminders and nudges to farmers to take action on a variety of issues that have to be addressed in a timely manner.
3. Enabling farmers to better understand market conditions to set prices for their produce and to manage inventory.

The idea of using AI solutions to provide decision support and other types of advisory in LMIC is itself not a novel idea[67,68]. However, the investments necessary to build such systems, ensure that they address the challenges particular to LMIC environments, provide interfaces that are usable by poorly educated users and across diverse sets of languages, and the maintenance and update burden have all severely limited the success of such an approach. With LLMs we have the benefit of a solution (e.g., a base model + a chat model) developed for general use as a starting point to build the LMIC specific solutions. These general models (i.e., those mentioned in Section 3 above) already have many of the core capabilities including user friendly conversation, emergent properties in reasoning, and training over a vast general corpus of relevant knowledge. Additionally, the LLMs themselves can help reduce the linguistic gap by using them as translators across many languages.

7. Challenges in Using LLMs in LMICs

Although LLMs have demonstrated impressive capabilities and seem naturally suited to address the aforementioned challenges faced in front-line care and advisory in multiple domains in LMICs, they cannot be directly applied to LMIC scenarios. There are three broad categories of LMIC specific challenges that must be addressed: (i) creation, (ii) deployment and scaling, and (iii) usage.

7.1. Challenges in creating and training the LLMs

The vast amount of data and information (“internet-scale”) used to create and train LLMs have led to the creation of autonomous and intelligent conversational agents and assistants that have shown significant emergent properties including human-like intelligence, but the data fed to these LLMs typically do not include local and contextual knowledge that may be essential to providing care in LMICs. These affect both base model capabilities and the fine-tuned models.

Base Model Capabilities:

The base model serves as a foundational layer, enabling the use of generic vocabulary and grammar to facilitate communication and retrieve universally applicable knowledge, such as basic math and anatomical terms, as well as fluency in conversation.

Despite its broad utility, the base models often fall short in addressing the unique challenges and conditions in LMICs. For instance, current training data used to create LLMS, even at "internet-scale", does not address the significant diversity in the variety of cultural practices related to diet, health management, and other general lifestyle aspects across regions, especially in rural areas of LMICs. These are not usually encoded in any form of document, especially in a digital form. Yet such factors can determine the type of care or treatment recommendation that may be accepted by the beneficiaries. Likewise, there may be time-honored methods in farming which have evolved over millennia that farmers may find beneficial and unwilling to change. These are not often recorded or available even at internet scale. Unless such knowledge is incorporated into the training of LLMs and integrated in the advice given by them, their practical value will be limited.

While it may be possible in principle to retrain base models to include data and information from LMIC contexts, it is not realistic or feasible to do so in practice. The effort and cost involved in doing so is prohibitively high, especially for LMICs. Hence, we rely on fine-tuning and specialization as the way to incorporate LMIC specific knowledge into LLMs.

Another challenge with using base models is that their emergent behaviors can also include undesirable/unexpected model behaviors. Such behaviors may include inappropriate or insensitive language, insensitivity to cultural norms, and unsuitable recommendations. The fine-tuning process is key to explicitly training the models to avoid such mishaps.

Fine-tuning and Specialization Capability in LMICs:

<u>Communication in local languages and in vernacular</u>

The vast linguistic diversity in the world, especially in LMICs poses a significant challenge. Languages spoken by communities tend to vary even over nearby regions, and variations in dialects can be even more prevalent. While many people tend to speak multiple languages, often they are more comfortable in using their native language/local dialect to describe their symptoms, needs and experiences. There is less variation in written language, but nonetheless it is significant. The greatest variations are often found among rural communities in poor countries. For the LLMs to be effectively useful, they need to be able to understand inputs from speakers in diverse languages and dialects and produce outputs in various languages.

Today, the base models have been predominantly and developed in English. The performance on various tasks such as summarization, information retrieval, etc. is proportionally poorer on low resource languages. While we may see the emergence of LLMs performing comparably in some of the major languages, it is unlikely to be feasible to do so in all languages, especially the "under-resourced" languages.

A natural solution is to develop translators along the lines of the effort being made at AI4Bharat [37] where LLMs themselves are used as a way to build translators between

English and many Indian Languages. It should be noted however, extending this approach to many dialects and vernacular communication is itself a significant programmatic undertaking.

Adaptation to socio-cultural norms

One of the greatest sources of communication mistrust is the variations in how people address each other, and how different simple expressions are understood in different communities. A simple example is the use of the term “please”. In some cultures, it may be considered impolite not to use this term, while in others it may be considered too formal and hence not relatable to the audience. Another common example is the use of the expression “I don’t care” as a response to being offered multiple choices. While in some cultures this simply implies that one does not mind whichever choice is given, in others it may imply a lack of concern and be seen as unfriendly. These types of linguistic nuances can play a significant role in creating (or preventing) a sense of equity and inclusion.

To ensure that the LLM adheres to socio-cultural norms, if the base model lacks guardrails optimized for LMICs while communicating with the LMIC audience, instruction-based fine-tuning can introduce these elements.

Contextual Knowledge

Local and contextual factors affect medical care and treatment as well as effective solutions in agriculture. For instance, the fact that red meat, a source of iron, is infrequently consumed in places like Gujarat, India, can be a factor in higher rates of anemia in pregnant women. In other examples, the risk of infection from various diseases often tends to be correlated with climatic and local environmental factors, and chemicals present in local water bodies may be an important factor in various health conditions. Likewise, farming practices tend to vary based on soil, climate, and even economic factors, and are often transmitted by word of mouth across generations.

Specialization is where the model becomes a subject matter expert. In LMIC applications, this can also be used to incorporate local and customized knowledge. Contextual knowledge can be used to enhance the model's knowledge, capability, and persona. A scalable way to frequently update this knowledge without retraining the model is called retrieval augmented generation (RAG). This approach enables the model to dynamically adapt to evolving information and adapt in real-time. For example, it could use rainfall records or the types of house dwellings (mud or thatch) as indicators for malaria risk. For instance, organizations like WHO have set a precedent by publishing specialized versions of their ANC Guideline documents targeting specific geographic challenges. In the domain of Agriculture, Digital Green has shown that farmer produced videos describing their best practices curated by them are widely popular and provide peer-to-peer knowledge sharing[38]. An LLM tailored to support farmers in LMICs can use such curated local information.

Hallucinations and Misinformation:

All AI based solutions that provide useful information also come with the risk of providing incorrect or false information. The LLMs are particularly vulnerable to this because they are trained in large corpuses of data which itself isn’t fully validated. As LLMs become

widely used, there is also the risk of them gaining false authority among the users, without full or proper authentication. In the context of LMIC users, this risk is amplified due to the fact that the users (e.g., CHWs) may not be equipped to challenge the guidance and recommendations provided by the LLMs.

Hallucination in the context of LLMs refers to instances where the model generates information that is not grounded in its training data or reality, effectively 'making up' details. Hallucinations represent a spectrum of failures, from mild to severe, that compromise the integrity of the content they generate. These failures can be categorized as follows:

- The risk of unwarranted certainty (Mild): The model presents information without indicating any uncertainty or ambiguity and issuing warnings about potential errors and alternative interpretations. For example, LLM might assert that “Implementing precision farming techniques will double your crop yield” without considering the specific environmental and individual farm management practices
- Failure to challenge the premise of the question (Moderate): The LLM does not scrutinize the premise of a question, potentially leading to the reinforcement of misinformation. An LLM could accept a flawed premise, such as "Since chocolate is a fruit, is it healthy?" without correcting that chocolate is derived from the cacao bean, which is a seed, not a fruit, thus potentially reinforcing misinformation.
- Caught in a lie (Severe): The LLM confidently generates verifiably false statements. For instance, an LLM might state with certainty that a particular medication is approved for a condition when it is not, which could have serious implications if believed by the user.
- Makes a wrong guess when forced to (Severe): When lacking information, the LLM may guess, leading to significant inaccuracies. LLM may fabricate a historical event's date when it lacks the information, leading to a significant distortion of historical facts.

These challenges with LLMs are widely recognized and addressing them is an active ongoing topic among the researchers and developers working in this area. It should be noted that LMIC usage settings and scenarios are particularly vulnerable to misinformation and misguidance due to already fragile access to knowledge.

One approach to address hallucination in LLM applications in LMICs is to build in additional safeguards in the form of human-in-the-loop solutions and human-supervision of the implementations at a system level. The human-in-the-loop approach involves incorporating human judgment into the decision-making process of the LLMs. Humans can provide real-time corrections and guidance, which can help in refining the model's responses. Human oversight also ensures that the output is verified and reliable, reducing the risk of hallucinations and improving the model's learning over time through active feedback.

Human supervision of the LLM implementation can be an important aspect of their deployment and management. We address this in the next section below.

7.2. Challenges in deploying, managing and scaling LLMs

LLMs are attractive because they can be an effective solution to addressing and possibly bridging many of the societal gaps that underlie the poor conditions in health and well-being, livelihood, and wealth, education, and other socioeconomic aspects in LMICs. However, there are significant challenges that can severely limit the deployment, management, and scaling of LLMs in these countries and environments. These include:

- Assumption of a homogeneous digital landscape and a robust digital infrastructure including the characteristics of a mostly available online network connectivity,
- Scaling the software development process through reusable software
- System level integration and change management (the transition from old to new methods that incorporate LLMs).

We now discuss these challenges in more detail.

Limitations due to poor digital infrastructure in LMICs

The practice of simplifying digital landscapes by categorizing countries as LMICs or non-LMICs is misleading. While non-LMICs often exhibit high tolerance levels that allow even inefficient software to function effectively, LMICs operate on narrow margins in aspects like Quality of Service and smartphone penetration. These constrained parameters in LMICs result in fragile, unreliable, and non-scalable software systems, with vulnerabilities often emerging during scale-up phases. Therefore, a one-size-fits-all approach is ill-suited, especially for LMICs. LLM implementations in LMICs therefore need to explicitly account for such fragility in the digital infrastructure and be tailored to the particular digital landscapes.

The most natural way for FLWs and beneficiaries in LMICs to interface with LLMs solutions is through a cell phone. Given the large size of LLMs the models themselves will often not be stored on a cell phone nor even a local server, but perhaps a more widely shared cloud storage resource, possibly at the National level. A smaller piece of software will be running on the cell phone, which frequently interfaces to the LLM stored on the cloud. However, the connectivity found in LMICs may be intermittent and unreliable, and such an interaction may not be smooth. We can expect delays in sending user inputs to the LLM and receiving the responses from the LLM.

LLM software solutions implemented in LMICs must be architected and configured in a manner that addresses the infrastructure constraints found locally. For example, LLM implementations in LMICs should preferably allow for a discontinuous conversation - for example, it would be beneficial if the solution a) is able to trickle data up to a server without any intervention by the user when online, and b) handle the resultant potential long latencies in conversation. Another example is using a mechanism such as “voice notes” found in common chat interfaces such as WhatsApp.

Scaling the software development process through reusable software

One of the key benefits of software approaches is the possibility to scale rapidly and at low cost. However, developing the next level of detail of this strategy should consider three factors:

1. It is important to leverage local software talent in individual LMICs to configure and maintain software, since this provides a pathway to scalable implementations throughout LMICs.
2. At the same time, it is also important to leverage software talent worldwide (from both LMICs and non-LIMCs) so that both reliable and world-class and newly developed core software capabilities are available to individual LMICs.
3. It is also important to incorporate specific requirements from local Governmental organizations and NGOs to be implemented and modified over time inexpensively and reliably.

The key is to design the solution in such a way that the software is reusable so that a wide range of talent in LMICs can build upon them. However, the approach cannot simply rely on functional open-source references since they may not have been designed with the three requirements above in mind. It also needs to be architected so that others can modify it easily so that training, debugging, and software maintenance on the ground are within scope of the widespread talent that is emerging throughout LMICs in software development for LLMs.

We recommend a coordinated oversight and development of baseline digital infrastructure that supports LLMs in all LMICs with input from stakeholders throughout all LMICs of interest. Baseline digital infrastructure may include voice or text data capture and transmission (either offline then online, or intermittent online), LLM server response (either offline then online, or intermittent online), data sharing (voice or otherwise) across different LLM use cases, as well as higher-level functions such as language translation. The development of such capabilities should not be repeated project by project, with highly variable results depending on the experience of the team.

There are precedents for this; they include the development of Open Health Stack [51] designed to solve some of the core challenges mentioned above in the realm of healthcare in LMICs. Extending the development approach, and potentially OHS itself to include LLMs could be a starting point to add core software infrastructure tools for LLMs. for LMICs.

System level integration and Change management

When performing insertion of digital technology in LMICs, there is often a focus on what a new digital technology is, as well as a focus on what it is replacing. Existing processes or technologies might be inefficient, but what they achieve may be mandated by the Government (e.g., report generation) and they may still have some efficacy. Likewise, there may also be prescribed limits on what sort of functions front-line workers are allowed to perform (e.g., in many countries CHWs are not permitted to perform medical diagnosis or prescribe treatment). New emerging capabilities may promise much greater performance or other benefits. However, typically there are resource and organizational

constraints that a worker or citizen needs to comply with to achieve their objective and simply asking them to replace an inefficient yet widely understood approach with a different approach is potentially risky for them professionally and perhaps personally also.

Moreover, new technologies (in this case LLMs), bring with them their own attendant risks. As noted in the previous section, LLMs carry the risk of misinformation and misguidance. While the natural growth and evolution of the technology can gradually reduce these risks over time, any system level integration of the solutions have to build in “safeguards” to mitigate against the failures due to these risks. These safeguards often involve human supervision and require a more gradual approach to the introduction of fully automated and unsupervised deployment. In other words, manual intervention should be part of a solution until sufficient confidence is achieved in the solution.

Change management between old and new processes is a necessary major task that directly addresses these issues. It is a means to achieve sustained adoption, which should be one goal of any impactful program. For example, an LLM use case may be replacing text-based look-up or an in-person help line. The triaging usage that we have discussed above may require revisions to the work protocols and responsibilities of different levels of personnel (CHWs, ANMs, Doctors). At least for a period of time, the old and new processes may have to work together.

In an example of how adoption was achieved in a paper-to-digital change management process in a healthcare application, technology was developed such that QR codes issued in a decentralized way were printed on paper that could then be stapled to paper records and scanned by the mobile phone of the worker. In this way there was both a paper and digital record available and indexed by each other, with the change management strategy of having the worker realize that the paper record had diminished value (and eventually no value) as their confidence in the digital record grew[52].

Additionally, it is important to ensure that every stakeholder from the ground (e.g. a CHW) up to senior Government officials realize an overall benefit by the addition of the new capability. We use the term “overall benefit” since for some workers (e.g. HCWs) in the transition process there may be some unavoidable drawbacks in their workflow (e.g. duplicating paper and digital processes), but with careful scoping of the program, these drawbacks can be vastly outweighed by benefits at other times in their workflow (e.g. avoiding end of month reporting, and the associated travel and frustrating working sessions to reconcile paper records). Similarly, technology insertions can be designed so that other stakeholders (e.g., clinic supervisors) realize a different benefit, for example up-to-date reports on clinic activity on a daily basis, which allows them to manage staff and consumables effectively without complaints, which are core parts of their responsibilities and personal key performance indicators and therefore are drivers of adoption of technologies.

In summary, a key requirement of the implementation approach is the smooth integration of the processes developed for LLM deployment and usage with incumbent processes. Success metrics need to be defined in order to ensure both the transition and the sustained adoption of a newly proposed LLM-based process, accounting for all stakeholders.

7.3. Challenges in using the LLMs

The success of LLM solutions, as in any software solution, not only depends on the capabilities of the system and the benefits it provides, but more importantly on the ease of its use and the comfort that the users feel while using the solution. Given the conversational nature of the LLM-based usage scenarios, user interaction becomes even more important. The challenges related to usage can be broadly divided into three broad categories: (i) ease of use, (ii) relevance of information provided, and (iii) ultimate benefit of the interactions and advice given. All three must be positively perceived by the user (e.g., FLW) as well as the end beneficiary (i.e., the patient, mother, farmer) to achieve sustained engagement with beneficial impact.

Ease of use requires LLMs to be fluent during the conversation in the language that is used for communication, including the understanding of vernacular. The conversational agent should appear human-like in that implicit information is understood by the agent, and the user's intent understood without having to explain it. It also includes handling disrupted conversations due to intermittent connectivity.

The user's perception of relevance of the information provided will depend on how well the LLM understands contextual factors and the constraints and opportunities that exist in the setting of the user. It also includes the agent being able to convince the user of the relevance in cases where the user may not fully understand the relevance.

Like all systems, the usage of LLMs will depend ultimately on the benefit that the user(s) perceive over time and the actual impact it has in enhancing health, well-being, wealth, education, and other aspects of life. Such impact will not be immediate, therefore, methodology for evaluating long term impact should be developed along the lines of what is typically done in social sector applications of technological and other solutions.

8. Evaluation of LLMs

There are three broad aspects to the evaluation of LLMs that are directly derived from the three challenges mentioned in Section 7. These are: (i) evaluation of LLM capabilities and the solutions they provide, (ii) evaluation of LLM deployment and availability, and (iii) evaluation of LLM usability.

8.1. Evaluation of LLM Capabilities: LMIC-Focused Metrics

As we noted in Section 4, while LLMs are being systematically evaluated even as they are being developed, the current evaluation frameworks may not fully capture the real-world nuances of large language models. Moreover, in the LMIC context, system and societal constraints add another layer of complexity to evaluations. These limitations highlight the need for a holistic, system-level assessment that contemplates the broader societal impact of these models.

The evaluation metrics and the frameworks must be adjusted and expanded to ensure key requirements for effectiveness of any solutions to LMIC are addressed. In the LMIC context, key aspects such as *Equity*, *Access*, and *Inclusion* (EAI) are also important. Further, what is perceived as necessary along these dimensions by the communities

themselves and what can be realistically achieved may vary significantly across cultures and geographies.

- **Equity:** Equity involves ensuring fairness in practices and policies. It means that everyone, regardless of their background, identity, or social status, has an equal opportunity to succeed. This often requires providing different resources or approaches to individuals or groups based on their specific needs or circumstances to address existing inequalities.
- **Access:** For a solution deployed in LMIC settings to have the necessary impact, access to the solution by all relevant users should be easy, consistent, and stable. However, universal access is often hard to achieve to perform a variety of factors including infrastructure limitation, language barriers, poor usability, low knowledge and skill among the users, and the lack of proper integration within the workflow resulting in poor or no support from the managers and administrators.
- **Inclusion:** Inclusion refers to creating an environment that actively welcomes, connects, and values all while harming none. It involves ensuring that everyone feels respected, valued, and supported, regardless of their identity or background. In an inclusive environment, everyone has an equal opportunity to participate, contribute, and express their perspectives.
- The three conventional LLM metrics that appear closest to addressing Equity, Access, an Inclusion are Fairness, Bias, and Toxicity (FBT). **Fairness**: In LLM evaluations, "fairness" is defined as the absence of any disparity in the task-specific accuracy of models across social groups. In other words, we can broadly define fairness as the absence of prejudice or preference for an individual or group based on their characteristics. For example, recommendation of opportunities should be neutral to a person's gender, ethnicity, or socio-economic status.
- **Bias:** Bias is often implicit and arises due to the lack of adequate representation of different groups or segments within the training data. Note that bias refers only to properties of what the model generates - i.e., there is no explicit relationship with the accuracy, or the specifics of a given task. Typical examples of bias involve implicit assumptions about variations in capabilities and limitations due to gender and racial differences (that are usually not true in real-life). Bias often arises due to lack of proper representation of different groups in the training data. Bias can lead to unfairness.
- **Toxicity:** Toxicity is defined as an instance-level property of text, encompassing concepts like hate speech, violent speech, and abusive language.

While success in Fairness, Bias and Toxicity metrics are essential for using LLMS in LMICs, they do not cover some aspects of EAI as discussed further below.

Equity is a broader societal concept than fairness or bias, specifically focusing on fairness in practices and policies that often require different resources or approaches to address existing inequalities. Fairness (HELM), however, is specific to machine learning models, focusing on the absence of disparities in the task-specific accuracy of models across different social groups. This does not ensure Equity in itself. Consideration of what may be necessary to overcome long standing obstacles to equity within the context of

usage within a local community would also have to be addressed. In others the solutions proposed must be effective and implementable.

Access goes beyond the solution itself. The ability of the end beneficiaries and the intermediaries (FLWs) to deploy and use the solution, to communicate with it orally in their own languages/dialects, as well as their ability to execute the recommendations made by the solution will also affect its access.

Inclusion is about actively creating an environment that values all individuals, regardless of their identity or background. While the reduction and removal of Bias and Toxicity can help improve inclusion they don't directly and actively address or ensure inclusion. There can be systemic limitations and who is allowed to do what within a community will have to be addressed.

To properly address LLM capabilities related to the EAI metrics requires significant programmatic effort. Such effort includes creating and identifying benchmark datasets, establishing criteria for success along these metrics and quantifying them, and creating a broadly accepted community wide evaluation process and implementing that process. Inputs on success criteria will have to solicited from all the stakeholders including representatives of the beneficiaries, government agencies responsible for the system level implementation, international bodies such as WHO, donors such as BMGF and USAID, NGOs and other organizations capable of implementation, technology providers and makers, as well as representatives of the broader technology community. The process should be done in a manner that it is naturally adopted by the community of users, makers, implementers, and policy makers.

8.2. Evaluation of LLM implementation and deployment

LLM deployment in the real world is governed and limited by a variety of infrastructural factors. In addition, when LLMs are deployed in real-world applications, they interact within a complex socio-technical environment. This environment introduces a multitude of factors that can significantly influence the model's performance and impact. Therefore, to fully understand and assess LLMs, it is crucial to consider their deployment within these broader contexts. These considerations are not just about optimizing model performance, but also about building user trust, ensuring ethical use, and maximizing the societal benefits of LLMs.

The technical aspects of LLMs form the backbone of their functionality. From deployment to continuous monitoring, these considerations ensure that the model operates efficiently and securely. They include:

Model's Deployment: Resource management, infrastructure cost optimization, and model performance. Neglecting these aspects can lead to suboptimal performance and increased costs.

Data Collection and Management: Covers the collection, storage, and handling of data used by the model. Poor data management can lead to security vulnerabilities and loss of user trust.

Model Development: Encompasses model versioning, prompt engineering, fine-tuning, creating chains, and domain-specific knowledge. These aspects influence the model's performance and adaptability to specific tasks.

Model Operation: Involves aspects like inference latency, scalability, security, customizability, interoperability, and maintainability. These factors are critical for the model's usability and security.

Continuous Monitoring: Ongoing tracking of the model's performance and behavior. This is essential for identifying and addressing issues in a timely manner.

8.3. Evaluation of LLM Usability

Understanding the user's perspective is key to creating a model that resonates with its audience. From user experience to data privacy, these considerations focus on building trust and meeting user needs:

User Experience: Building trust, defining the scope of complex actions, and ensuring accessibility. These factors significantly influence user satisfaction and trust in the model.

User Feedback: Incorporates human feedback loops to improve the model's performance and behavior. This is crucial for ensuring the model meets user needs.

User Data Privacy: Ensures the privacy and security of user data. Neglecting this can lead to legal issues and loss of user trust.

Costs: Considers the financial implications for the user. High costs can hinder the model's accessibility and adoption.

PART III - LLM SOLUTIONS IN LMIC – ILLUSTRATIVE EXAMPLES

9. Illustrative Use case examples

One of the primary objectives of this document is to assess the potential of LLMs for LMIC usage and applications, establish the requirements on LLMs for such usage and propose an approach to a more systematic and comprehensive evaluation. While a complete evaluation is both ambitious and open-ended and beyond the scope of this document, we will use a few examples to illustrate how we might go about such an undertaking.

We consider three examples of using LLMs in LMIC settings: two in the health domain and one in agriculture. Note that these examples are simply illustrative, and it is not our intention to suggest that this set of example use-cases chosen is complete and comprehensive. However, it is our hope that these examples serve to illustrate how one might go about defining the core capabilities required of LLMS and some of the key metrics assessing and evaluating LLMs for LMIC applications.

Use Case Outline

Each Use Case follows the following structure:

- Description
- Problem Statement
- Solution Abstract
- Systemic Constraints
- Scope and Boundaries
- Solution Design
- Data Management
- LLM Selection and Adaptation
- LLM Specific Metrics

<u>Use Case 1: Detection and Management of High-Risk Pregnancy Conditions</u>

a) **Description:** The delivery of primary health care services in LMICs rely heavily on Community Health Workers (CHW) especially in remote and underserved regions. Maternal and child health (MNCH) is a key area of their focus. For example, the primary healthcare system in India relies on Accredited Social Health Activists (ASHA) who are community health workers. ASHAs follow a well-defined work routine to support mothers during pregnancy as well as during the early period of the infant's life. The care is provided usually through home-visits.

b) **Problem Statement:** The early detection and management of high-risk pregnancy conditions is of high priority within MNCH in LMICs. Early and effective detection and proper management is necessary to reduce maternal and neonatal mortality and

morbidity. The effectiveness of ASHAs in providing such care is often limited by a lack of immediate access to up-to-date and contextual medical knowledge. This is compounded by inadequate supervision and support to ASHAs, resulting in low coverage of selected MNCH services and care received by complicated cases.[42,43]

c) **Solution Abstract:** A potential solution to the above problem is to integrate Large Language Models (LLMs) into an interactive chat platform to serve as a knowledge assistant for community health workers. Such a tool will offer real-time, evidence-based medical guidance to ASHAs for assessing pregnancy risk. In addition to using standard sources of knowledge such as WHO guidelines, the LLMs can be tuned to leverage information that is aligned with local health guidelines and best practices and take into consideration the local contextualized information about living conditions, lifestyles, diet and nutrition, cultural practices, and restrictions as well as mother specific information. A human-in-the-loop should be available for verification and additional support.

d) **Systemic Constraints:**

Local Relevance: Medical guidance must align with local health guidelines, cultural norms, and available resources. For example, advice for a vegetarian anemic patient from rural India would differ from that for a non-vegetarian patient.

Usability: The platform must be user-friendly and accommodate varying literacy and technological proficiency levels. A preference for local languages and voice-based interfaces is common.

Data Privacy: Compliance with healthcare data regulations is essential to ensure the security of patient and healthcare worker data.

e) **Scope and Boundaries:** The LLM will act as a knowledge assistant and is not intended to replace formal medical training or consultations. It will assist ASHAs in accessing medical information necessary for determining pregnancy risks, guide the ASHAs in performing the necessary medical tests and take measurements, and possibly triaging. Critical decisions about the actual kind and level of risk as well as the recommended treatment, however, will still involve experienced medical professionals. A human expert will always be available as a fallback should the LLM's response be inadequate.

f) **Solution Design:**

Users and Stakeholders

Primary Users: ASHA workers in the Indian primary healthcare system.

Secondary Stakeholders: Auxiliary Nurse Midwives (ANMs) who supervise ASHAs, Doctors, Healthcare administrators, local governments, and healthcare-focused NGOs.

Workflow and Human-in-the-Loop Design

1. User Query: The ASHA initiates a query via a text and/or voice-enabled interface.

2. Initial Response: The LLM provides a preliminary answer, rooted in a curated and contextual medical knowledge base to avoid hallucinations and biases.
3. A more detailed conversation ensues.
4. The LLM may recommend that various measurements are taken from the mother and that tests are performed. The ASHA performs those tests that are possible within her available resources and conditions and seek alternatives when something is not possible.
5. Human Verification: Queries with low model confidence in critical or complex cases are flagged and sent to medical experts for verification.
6. Feedback Loop: Users can provide feedback on the utility and accuracy of the information, aiding the model's ongoing improvement.

Functional Requirements

- Real-time response to user queries
- Offline functionality
- Incorporation of local contextualized information.
- Support for multiple local languages.
- Voice-to-text and text-to-voice capabilities, along with machine translation from local languages to English.
- A user-friendly interface.

Technical Requirements

- Low-latency query processing for real-time responses.
- Local caching for frequently accessed information.
- Compatibility with low-end mobile devices and tablets.

g) Data Management:

Data Collection: Data will be sourced from various channels including medical literature, Asha Handbooks, ICMR guidelines, and other trusted and expert-verified sources. This curated knowledge corpus defines the scope of the LLM's responses and will also include insights from local medical professionals for regional relevance.

Data Cleaning and Labeling: Strict data cleaning protocols will be in place to ensure data integrity. A human-in-the-loop system will be used for data labeling to assure the quality of training data.

Data Privacy and Governance: The platform will comply with both local and international data privacy regulations. Strong encryption protocols will be implemented, and regular audits will be conducted to ensure data security. Data will be stored on local servers to align with data sovereignty laws and improve access speed in areas with low connectivity.

h) LLM Selection and Adaptation

Core LLM Component:

The primary focus will be on general-purpose Large Language Models (LLMs) that can be specialized to disseminate medical information, particularly in relation to high-risk pregnancy conditions.

Level of Adaptation of LLMs Required:

Prompting: The initial adaptation stage employs specific input sequences or questions to guide the LLM's output. This is the least resource-intensive approach and is suitable for addressing basic medical queries.

Retrieval Augmented Generation (RAG): In this moderately resource-intensive level, the LLM interfaces with external databases of the protocol documents, FAQs and other source material will be accessed in real-time, providing contextually accurate, and detailed responses.

Fine-Tuning: This is the most resource-intensive level and involves using a curated dataset and computational power to tailor the LLM for very specific tasks, such as local medical practices, contextual language and cultural context.

Base Model: In LMIC settings, typically, the cost of training a base model from scratch is currently prohibitively high. Hence, all approaches will have to rely on existing base models. Using the base model without any adaptation is theoretically possible but unlikely in practice. Moreover, the limited availability of high-quality, localized medical data makes it less feasible to train a reliable and effective base model.

The choice of adaptation level will be iterative. The system will start with simple prompting techniques, move on to few-shot prompting, advance to employing RAG, and potentially arrive at human-annotated fine-tuning if necessary. Given the low-resource settings, retraining the base model is considered unlikely, making this approach cost-effective.

Software Tools and Ecosystem:

Core Machine Learning Libraries: TensorFlow, PyTorch, Langchain,

Vector Databases: Pinecone, ChromaDB

Compute/Cloud Infrastructure: AWS, GCP, Azure

LLM Configuration Choice:

In the use case where ASHA workers answer questions that are factual and evolving (due to changes in protocols, new medications, government programs, and updated links to next steps), then an LLM configured to use RAG (Retrieval Augmented Generation) is advisable. This is because RAG focuses on retrieval of information from a fixed corpus of information which greatly reduces the likelihood of hallucinatory or incorrect responses. This can be compared to an LLM configured without RAG where responses may be more creative but less predictable. The LLM could be either open or closed, fine-tuned or not. In this use case which involves searching and answering factual questions, a 'creative' model is not preferred.

i) **LLM Specific Metrics**

Apart from the standard LLM metrics (e.g., the 7 metrics considered by the HELM framework), in this section we consider the particular metrics relevant for this LMIC centric use case.

- Linguistic and Dialectal Coverage: Measures the LLM's ability to understand and converse in local languages and dialects, which are often deeply connected to cultural, caste, and socio-economic identities.
- Accessibility and Usability:
- Literacy-Adaptive Helpfulness: Tailors the model's helpfulness to the literacy rate of the healthcare workers, ensuring that even those with basic literacy levels can effectively use the model.
- Digital Literacy: Evaluates the ease of use for healthcare workers with different levels of technological proficiency. Younger health workers are more likely to be digitally literate, making this an essential factor for equitable access.
- Cultural Sensitivity: Assesses the appropriateness of advice in the context of local traditions and beliefs, ensuring that medical guidance is culturally sensitive.
- Contextual Resource Sensitivity: Assesses whether the medical guidance considers the availability of resources and feasibility of recommendations based on the local context. For example, the model should avoid suggesting expensive procedures like CT scans if they are inaccessible or unaffordable to the user. Instead, it should recommend viable options aligned with the constraints of the local healthcare system and the individual's circumstances.
- Gender Sensitivity: Given that both the beneficiaries (the mothers) and the healthcare workers in this use case are exclusively women, this metric assesses the model's ability to address the unique medical and psychosocial needs of women.

By focusing on these multi-dimensional metrics, we aim to ensure that the LLM is not just clinically effective but also socially responsible. Meanwhile, the DEI metrics focus on making the technology accessible and relevant to the diverse range of healthcare workers in LMICs, keeping in mind that in this specific use case, both the beneficiaries and the healthcare providers are women, making certain aspects of DEI to be more nuanced.

Use Case 2: LLM-assisted Digital Transformation of Healthcare Workflows in Nigeria

a) Description

The Community Health Influencers, Promoters and Services (CHIPS) program in Nigeria is designed to encourage and support local households to seek PHC services at clinics and outreach programs.[44]

The program also helps clients to get necessary PHC services by bringing these services closer to households through home visits by CHIPS Agents, especially in rural underserved regions.

Once patients reach the clinic, then it is important that any unusual clinical events are flagged, and messaging created to encourage the patient to adhere to their care plan, including returning to the clinic.

b) Problem Statement

CHIPS agents are typically non-medical workers that travel door to door with no medical support easily available, yet need enough information to identify that a patient may need service across a wide domain of healthcare issues (ANC, malaria, STDs, etc.). They also need to have enough information and credibility to encourage the patient to seek the relevant PHC services.

In addition, even if the patient reaches the clinic, existing solutions typically lack the ability to detect any unusual symptoms and trends, while on a tactical level they provide poor general clinic management. This results in poor disease surveillance and prevents effective response to disease outbreaks. Finally, even if a patient reaches a clinic and a care-plan is identified, then there is often little patient engagement which frequently results in loss to follow-up.

c) Solution Abstract

A natural approach to address the situation in Nigeria and to achieve improved outcomes is through the innovative use of LLMs accessed on digital mobile phones to transform healthcare workflows. Non-LLM solutions using lower-end digital mobile phones even with poor connectivity have been successfully deployed in Nigeria https://www.iprdsolutions.com/impact-health. It is natural to add LLM capability to such digital solutions.

Such an implementation could consist of:

1. Summarization of verbal and textual conversations between HCW and patient in clinic or community setting, as well as summarization of structured encounter data to date.
2. Use of natural language querying of collected data to i) identify relevant educational material or specific guidance to the CHIPS worker on issues while they are in the field, and ii) detect and anticipate unusual clinical events when a patient reaches the clinic.
3. Use of creative and inspired messaging using LLMs to improve patients' engagement in their own health outcome e.g., adherence and clinic recall.

d) Systemic Constraints

Local Relevance: A CHIPS worker has to identify the possibility of a range of health conditions in their patients, from pregnancy to malaria to STDs. Medical guidance must align with local health guidelines, cultural norms, and available resources. The types of patient engagement should be highly scalable, and proven communication and engagement methods should be used where identified.

Usability: The platform must be user-friendly and accommodate varying literacy and technological proficiency levels. A preference for local languages and voice-based interfaces is common.

Data Privacy: Compliance with healthcare data regulations is essential to ensure the security of patient and healthcare worker data.

e) Scope and Boundaries

The LLM will act as a knowledge assistant and is not intended to replace formal medical training or consultations. It will assist CHIPS workers in accessing medical information necessary to determine if the patient needs to seek care of a particular type. The LLM can also assist in encouraging the patient to follow the advice of the CHIPS worker. It is not necessary for the LLM to make a diagnosis, but only alert that seeking care is advised for given reasons. Within the clinic, an LLM can be used to follow the care-plan of the patient, identify anomalies, and create inspiring messaging to encourage adherence to the care-plan. The LLM does not need to perform diagnosis.

f) Solution Design

Users and Stakeholders:

Primary Users: a) CHIPS workers and b) HCWs, in the Nigerian healthcare system

Secondary Stakeholders: Supervisory HCWs, Doctors, Healthcare administrators, local governments, and healthcare-focused NGOs.

CHIPS Workflow and Human-in-the-Loop Design

- User Query: The CHIPS worker initiates a query via a text and/or voice-enabled interface.
- Initial Response: The LLM provides a preliminary answer, rooted in a curated and contextual medical knowledge base to avoid hallucinations and biases.
- A more detailed conversation ensues.
- The LLM may recommend performing basic tests that are available to the CHIPS worker (e.g., a malaria RDT test). The CHIPs worker performs the tests that are possible within his or her available resources and conditions and seeks alternatives when something is not possible.
- Human Verification: Queries with low model confidence in critical or complex cases are flagged and sent to medical experts for verification.
- Feedback Loop: Users can provide feedback on the utility and accuracy of the information, aiding the model's ongoing improvement.

In-clinic HCW Workflow and Human-in-the-Loop Design

- User Query: The query is preprogrammed to i) summarize the conversation between the HCW and patient, ii) identify unusual clinical aspects in the longitudinal health record, and iii) define a concise message to engage the patient to maintain their care plan, which may include drug adherence or a return to the clinic.
- Initial Response: The LLM uses the contextual information from the CHIPS agent that referred the patient to the clinic, as well as any tests that were done (e.g., an RDT result). The LLM also uses the conversation between the HCW and the patient.
- Human Verification: Queries with low model confidence in critical or complex cases are flagged and sent to medical experts for verification.
- Feedback Loop: Users can provide feedback on the utility and accuracy of the information, aiding the model's ongoing improvement.

Functional Requirements

- Summarize patients' health history by creating succinct actionable sentences about the patient to make it easy to manage repeat visits.
- Create personalized nudges to increase patients' engagement in health outcomes, e.g., personalized text messages with call to action and messaging to draw attention.
- Offline functionality for CHIPS worker use-cases, and for identifying unusual clinical aspects in the longitudinal health record (so that the HCW is alerted while the patient is present).
- Online functionality is suitable for summarization and for forming patient engagement messaging since this can be sent a few days later once the phone of a HCW has been synchronized with a server.
- Incorporation of local contextualized information.
- Support for multiple local languages.
- Voice-to-text and text-to-voice capabilities, along with machine translation from local languages to English.
- A user-friendly interface.

Technical Requirements:

- Native language speech to text to generate native language transcripts.
- Translation from Native language to English to generate English transcripts.
- Mapping form fields (questionnaire items) to questions for LLM
- LLM for extracting answers from English transcripts

- Module for capturing feedback extracted answers.
- LLM for summarizing patient history to draw HCWs attention to the most important actionable information.
- Module for capturing feedback on generated summaries.
- LLM for generating personalized nudges for patients based on their recent encounters.
- Mechanisms (e.g., A/B testing) for feedback/control on nudges

g) Data Management

Data Collection: Data will be sourced from various channels including medical literature, CHIPS handbooks, WHO guidelines (including country-specific versions), and other trusted and expert-verified sources. This curated knowledge corpus defines the scope of the LLM's responses and will also include insights from local medical professionals for regional relevance.

Data Cleaning and Labeling: Strict data cleaning protocols will be in place to ensure data integrity. A human-in-the-loop system will be used for data labeling to assure the quality of training data.

Data Privacy and Governance: The platform will comply with both local and international data privacy regulations. Strong encryption protocols will be implemented, and regular audits will be conducted to ensure data security. Data will be stored on local servers to align with data sovereignty laws and improve access speed in areas with low connectivity.

h) LLM Selection and Adaptation

Core LLM Components:

The core LLM components required are:

- Native Speech to text
- Native language to English translation
- LLM for extraction
- LLM for summarization
- LLM for generating personalized nudges.

For the CHIPS worker, the primary focus will be on general-purpose Large Language Models (LLMs) that can be specialized for identifying potential healthcare issues across a broad spectrum of diseases, particularly with a focus on the diseases / conditions that are most prevalent in that geography. The LLM here needs to work in an offline use case.

For the HCW in the clinic, there is typically already a down-selection of the health area being addressed (e.g., ANC, STD), and therefore the LLM can focus primarily on this limited domain. For the immediate flagging of health anomalies, then the LLM needs to work offline. For summarizing the HCW/patient interaction, and for creating engaging

messaging to avoid loss to follow-up or adherence, then an online-only LLM capability is suitable since these messages can wait until the HCW has synchronized their phone a few days later.

Level of Adaptation of LLMs Required:

1. Prompting: The initial adaptation stage employs specific input sequences or questions to guide the LLM's output. This is the least resource-intensive approach and is suitable for addressing basic medical queries. This may be sufficient for many rudimentary cases that a CHIPS worker has to address.
2. Retrieval Augmented Generation (RAG): In this moderately resource-intensive level, the LLM interfaces with external databases of the protocol documents, FAQs and other source material will be retrieved to provide contextually accurate, and detailed responses. In the offline CHIPS worker use case, a small subset of these external databases may be curated and downloaded onto the phone of the CHIPS worker. The subset can potentially be chosen by the manager of the CHIPS worker, by selecting topic areas that are most prevalent in their working region.
3. Fine-Tuning: This is the most resource-intensive level and involves using a curated dataset and computational power to tailor the LLM for very specific tasks, such as local medical practices, local language, and cultural context. This fine-tuning step may not be possible in the offline use cases but is possible in the online use cases (summarization and creating engaging messaging). Fine-tuning in an online mode can also be used to refine any initial flagging of possible diseases or outbreaks that were created in an offline mode.
4. Base Model: In LMIC settings, the cost of training a base model from scratch is prohibitively high. Hence, all approaches will have to rely on existing base models. Using the base model without any adaptation is theoretically possible but unlikely in practice. Moreover, the limited availability of high-quality, localized medical data makes it less feasible to train a reliable and effective base model.

The choice of adaptation level will be iterative. The system will start with simple prompting techniques, move on to few-shot prompting, advance to employing RAG, and potentially arrive at human-annotated fine-tuning if necessary. Given a low-resource setting, retraining the base model is considered unlikely, making this approach cost-effective.

Software Tools and Ecosystem:

- Core Machine Learning Libraries: TensorFlow, PyTorch, LangChain,
- Vector Databases: Pinecone, ChromaDB
- Compute/Cloud Infrastructure: AWS, GCP, Azure

LLM Configuration Choice:

In the use case where CHIPS workers have to retrieve factual information from a corpus of guidelines, then an LLM configured to use RAG (Retrieval Augmented Generation) is

advisable. This is because RAG focuses on retrieval of information from a fixed corpus of information which greatly reduces the likelihood of hallucinatory or incorrect responses.

In the use case where data is being summarized a RAG-enabled LLM may constrain the response to de-emphasize unusual information, although this may be mitigated if RAG is configured with a wide range of case study examples. The use of RAG where possible would limit hallucinatory summarizations.

In the use case where the model needs to generate personalized and persuasive text to reduce loss-to-follow up, a fine-tuned model is preferred. This is because the model's intent must be trained to operate in a generative mode while being sensitive to what is persuasive in that geography and use case, e.g., providing a next visit date with the positive and negative outcomes that can potentially happen if the visit is kept or not, given the context of the patient's current condition and geographical and other contextual information.

Connectivity Solutions:

For areas with low or intermittent connectivity, a localized, smaller model will reside on community health workers' smartphones as a fallback option. Additionally, cached responses for common queries will be made available offline.

i) LLM Specific Metrics

Apart from the standard LLM metrics (e.g., the 7 metrics considered by the HELM framework), in this section we consider the particular metrics relevant for this LMIC centric use case.

- Linguistic and Dialectal Coverage: Measures the Solution's ability to understand and converse in local languages and dialects, which are often deeply connected to cultural, racial, and socio-economic identities. Coverage for traditional languages like Hausa, Yoruba, Igbo as well as pidgin which is popular amongst young people, should be evaluated.
- Accessibility and Usability
- Literacy-Adaptive Helpfulness: Tailors the solution's helpfulness to the literacy rate of the healthcare workers, ensuring that even those with basic literacy levels can effectively use the solution, while those with more advanced literacy skills can be helped more (for example, a faster workflow)
- Digital Literacy: Evaluates the ease of use for healthcare workers with different levels of technological proficiency. Younger health workers are more likely to be digitally literate, making this an essential factor for equitable access.
- Cultural Sensitivity: Assesses the appropriateness of advice in the context of local traditions and beliefs, ensuring that medical guidance is culturally sensitive.
- Contextual Resource Sensitivity: Assesses whether the guidance considers the availability of resources and feasibility of recommendations based on the local context. For example, the model should potentially avoid suggesting tests like an

ultrasound or uncommon RDT tests, which are either unavailable or against standard operating procedures in particular cultures/geographies. Instead, it should recommend viable options aligned with the individual's circumstances and the tests that a CHIP agent is supplied with.

- Gender Sensitivity: Given that both the beneficiaries (the mothers) and the healthcare workers in this use case are exclusively women, this metric assesses the model's ability to address the unique medical and psychosocial needs of women.

By focusing on these multi-dimensional metrics, we aim to ensure that the LLM is not just clinically effective but also socially responsible.

Use Case 3: Farmer.CHAT

Increase agricultural productivity through LLM-based advisory to farmers and Extension Agents

a) Description

Traditionally, farmers receive support from public extension systems, which typically deliver advisory via field demonstrations, or in-person training facilitated by frontline workers also known as Extension Agents (EA). These practices are not cost-effective at scale, vary greatly in quality, and largely disseminate generalized top-down driven advisory, with major gaps in and lack of consideration for locally relevant extension messages. AI powered chatbots that are based on LLMs can help complement the EA system by integrating local and contextualized knowledge and information.

b) Problem Statement

In addition to the problems described above, current systems face other challenges that include limited reach, language and communication barriers, limited private sector involvement, and lack of evaluation mechanisms.

Leveraging peer-to-peer learning and existing social networks, Digital Green has cost-effectively amplified the reach and effectiveness of partners through use of videos produced by and for farmers[38]. Farmers have been accessing these videos through apps such as WhatsApp & YouTube. (Digital Green's YouTube channel has 81mn+ views.)

While this approach has made significant contributions to the improvement of farm practices that are ecologically more sustainable and economically productive, integrating this solution with the extension worker program together with the global knowledge that is available would lead to a more complete and fully scalable solution. In addition, these videos need to be integrated with other datasets like farmer and extension agent registries which include location, commodities of interest, and other types of useful information.

c) Solution Abstract

A natural solution is to build an AI-powered chatbot to understand the needs, preferences, and values of the farmers and increase their agency by engaging them as primary users the of bot, but in addition allow Extension Agents also to be its users so that they can provide the proper interface to the EA system and the relevant government programs. In

fact, Digital Green[38] has already been working with its partners to prototype an LLM based conversational agent called Farmer.CHAT [45,46].

The discussion below is based on this living example, which will enable increasing extension agent and farmer engagement (focused on women and low literacy and low digital skill populations), the integration of government advisory information together with local knowledge, and the inclusion of current market data to provide economic guidance.

d) Systemic Constraints

Local Relevance: Farming conditions and practices vary significantly across regions; hence local community knowledge needs to be integrated.

Usability: The platform must be user-friendly and accommodate varying literacy and technological proficiency levels. A preference for local languages and voice-based interfaces is common. Many farmers are women - they often do not have access to higher-end cell phones and may only have a simpler "feature phone".

Data Privacy: There is a need to protect farmer's identity, location and other personal information. Government regulations and protocols tend to vary across countries and regions and have to be respected.

e) Scope and Boundaries

The LLM cannot replace the EA advisory system, but rather should be perceived as a complement and an extension to that system. It should therefore be usable both by farmers and EAs. However, the way the two groups of people use the system is likely to be different in significant ways. The system will also need to work with government systems that include EA registries, for example, to ensure adaptation and long-term sustainability.

f) Solution Design

Conversational agents based on LLMs have the potential to provide timely and contextualized advice to farmers on a variety of issues. A solution can be conceived along the following lines:

- Extension workers have access to a multilingual LLM chat agent that helps them answer farmers' questions and concerns about a variety of topics: e.g., soil management, pest controls, crop diseases, market conditions, and other factors.
- This LLM-based chat agent leverages a knowledge base that is composed of Digital Green videos, Kisan Call Center logs, to the local and state govt advisory memo system, relevant scientific knowledge from agriculture institutions and community knowledge from other similar farmers.

Users and Stakeholders:

Primary Users: Extension workers and Farmers

Secondary Stakeholders: Agricultural administrators, companies developing pest management solutions, and NGOs supporting rural farmers in LMICs.

Workflow and Human-in-the-Loop Design

1. User Query: Farmers or Extension workers ask a question about the current issue of concern: e.g., pest management. This question is likely to be sent as “code mixed” text, i.e. a mix of Indic local-language terms, often phonetically spelled in Roman characters OR as a WhatsApp audio message.
2. Initial Response: The LLM provides a preliminary answer, rooted in a curated knowledge base as described in the following data section: e.g., in the case of pest management, the LLM may provide information about current infestation outbreaks and the symptoms to look for.
3. A conversation ensues wherein the farmer/extension worker may provide additional contextual information.: e.g., such information may include photos of the crops.
4. The LLM may recommend various tests and trials of products for the farmer to conduct.
5. Human in the loop: Farmer or the Extension worker have a way to escalate unresolved queries to the next level of expertise.
6. Feedback: Every LLM response is shown with

Functional Requirements:

- Support for multiple local languages.
- Incorporation of local and contextual knowledge from farmers.
- Voice-to-text and text-to-voice capabilities, along with machine translation from local languages to English.
- A user-friendly interface for beneficiaries, ideally using tools such as WhatsApp, IVR or Telegram that they already use.
- A user-friendly interface for Expert content curators of the knowledge base - e.g., Google Sheets and Docs to edit video transcripts (and resulting synthetic data), FAQs, and other documents.

Technical Requirements:

- Cached LLM embeddings of knowledge base documents (Redis is an example capability), with local caching for frequently accessed information.
- Compatibility with low-end mobile devices and tablets for beneficiaries.
- Automated testing abilities to assess how changes to prompts, knowledge base documents, models, LLMs, and other components affect the quality of output.
- Feedback mechanisms
- Text based LLM prompts to:
    - Summarize a user's chat history for accurate vectorDB querying.
    - Reasoning capabilities with large context windows to incorporate matching document snippets, reasoning workflows, real-time data (e.g. whether), user profile information and conversation history to create accurate advisory.
    - Conversational analysis .
- Ability for the LLM to also analyze images from the field for diagnosis
- Robust translation systems that can be modified for specific use cases e.g. "Chili" translates to both capsicum and hot chili peppers in the dominant English Internet and hence, LLMs often confuse chili for capsicum in knowledge base searches (there are whole categories of chilis in India each with separate names). We need systems where such confusion can be identified and corrected.

g) Data Management

Dataset Curation:

- Collection of the relevant knowledge based from the agriculture institutes and local and state govts about the pest management protocols and workflows which would

include necessary recommendations like the schedule and the type of pesticide usage, seed and fertilizer selection, climatic conditions based advisories and pest prevention guidelines

- In addition to static knowledge, real-time weather and local pest attacks are also accessible to the LLM model to provide accurate and contextualized advisory.
- To increase inclusivity a lot of background science of the crops would also be added for users to probe the model deeper and learn about it
- Curation of content from community knowledge sharing platforms like Digital Green would also be part of the knowledge corpus.
- In Farmer.CHAT the curated community knowledge is translated to English and stored in the form of Google docs which is pulled into a vectorDB with each query. It also allows the knowledge base to be updated in real-time.

Data Cleaning and Labeling:

- With help from the experts, contradictory and stale information will be removed from the knowledge corpus.
- The local language knowledge corpus should be translated to English.
- A comprehensive set of questions and answers will be created to evaluate the model's performance and other relevant metrics described below.

Data Privacy and Governance

The platform will comply with both local and international data privacy regulations. Strong encryption protocols will be implemented, and regular audits will be conducted to ensure data security. Data will be stored on local servers to align with data sovereignty laws and improve access speed in areas with low connectivity.

h) LLM Selection and Adaptation

LLM Capabilities and Components:

The core LLM components required are:

- Native Speech to text
- Native language to English translation
- Incorporation of native/local content
- LLM for summarization
- LLM for generating personalized nudges.
- Contextualized image recognition, using multi-modal LLMS.
- Report generation (useful for Extension Agents)

- An abstraction system such that new AI models can be constantly tested in the system and swapped in if they perform better.

Level of Adaptation of LLMs Required:

Prompting:

- Given the use case is about hyper-local pest management issues, it is unlikely that the base model or the general-purpose chat model would have all the relevant information.
- Hence for this use case simply prompting the model will not be sufficient.

RAG:

- This use case is ideal for a RAG-like system where we can control the knowledge base of the model and force it to cite it's answer which increases the reliability of the system.
- This also allows us to update the knowledge base without costly updates to the model.
- One of the limitations of this approach is that for each response from the model the knowledge it could 'account' for is limited to its context width which is 32K token for the latest GPT-4 model. Farmer.CHAT works around this by converting the local knowledge corpus into a VectorDB, then querying it to get back document snippets and pack them along with any other profile data, weather data, conversation history, and other factors into the context window.

Fine tuning:

- Fine Tuning in this case would make sense as we could embed the knowledge corpus into the model. Unlike the RAG system there is no limit to the knowledge the model can consider responding to a query.
- Typically, the unchanging knowledge would be the universal concepts of pest management, while the scientific discoveries would come from a preferred choice of the knowledge corpus that can be embedded into the model.
- In addition, fine tuning can be beneficial in reducing culturally inappropriate responses by training on an established query and response dataset from humans.
- Although this is more resource intensive it would greatly increase the models' focus on this narrow domain, thereby increasing its usefulness while reducing irrelevant and incorrect responses.
- In addition to fine-tuning the LLM we would also consider fine-tuning the image model to understand pest management related images like the damaged leaf, soil, pest images and satellite images.

LLM Configuration Choice:

In the use case where ASHA workers answer questions that are factual and evolving (due to changes in protocols, new medications, government programs, and updated links to

next steps), then an LLM configured to use RAG (Retrieval Augmented Generation) is advisable. This is because RAG focuses on retrieval of information from a fixed corpus of information which greatly reduces the likelihood of hallucinatory or incorrect responses.

i) **LLM Specific Metrics**

Apart from the standard LLM metrics (e.g., the 7 metrics considered by the HELM framework), in this section we consider the metrics that are particularly relevant for this LMIC centric use case.

Linguistic and Dialectal Coverage: Measures the LLM's ability to understand and converse in local languages and dialects, which are often deeply connected to cultural, caste, and socio-economic identities. We recommend that we also create an easy-to-use measurement infrastructure, wherein organizations can bring their own datasets - e.g., 100 recordings of rural Bhojpuri asking questions encoded as WhatsApp audio notes - and we can easily assess the performance of the latest AI Models on their particular dataset.

Accessibility and Usability:

- Literacy-Adaptive Helpfulness: Tailors the model's helpfulness to the literacy rate of the farmers and extension workers, ensuring that even those with basic literacy levels can effectively use the model.
- Digital Literacy: Evaluates the ease of use for farmers and extension workers with different levels of technological proficiency. The younger generation are likely to be more digitally literate, making this an essential factor for equitable access. For low digital, non-smartphone users, these conversational bots could be made more usable via language interactive voice response (IVR) interfaces.
- Cultural Sensitivity: Assesses the appropriateness of advice in the context of local traditions, beliefs, and farming practices, ensuring that the guidance given is culturally sensitive.
- Contextual Resource Sensitivity: Assesses whether the guidance considers the availability of resources and feasibility of recommendations based on the local context.
- Gender Sensitivity: Women farmers tend to face a variety of systemic and cultural issues in addition to the challenges faced by male farmers. For example, in some cases women farmers may not have a smartphone and instead rely on a traditional “feature phone”. This raises the need for IVR/SMS type interfaces. There may also be cultural rules and restrictions on the functions they perform, and how they are able to interact with social media and peer groups. A key metric would be to measure the LLM’s ability to address and handle gender-related usage and usability issues.

By focusing on these multi-dimensional metrics, we aim to ensure that the LLM is not just effective but also socially responsible.

PART IV - SUMMARY AND RECOMMENDATIONS

10. Summary

In this document we have attempted to provide an insight into the rapidly emerging breakthrough technologies in Artificial Intelligence called Large Language Models (LLMs) and to understand how their vast potential can be extended to enhance the lives of poor and underserved communities in Low and Middle-Income Countries LMICs. We have also described a framework to assess their use and application in LMIC conditions and constraints.

Evaluating and understanding Large Language Models requires a multifaceted approach across research and real-world contexts. On the research side, comparative assessment using diverse models, metrics, benchmarks, and scenarios provides valuable insights into model capabilities and limitations. However, the real-world deployment introduces many additional technical, user-centric, ethical, legal, societal, and environmental considerations.

To build trust and realize the full potential of LLMs, it is crucial to evaluate them rigorously both in controlled settings and real-world systems. The goal should be not just raw performance but maximizing usefulness while ensuring responsible and ethical use. This requires collaborative efforts between researchers, developers, users, and broader society. Evaluation and governance frameworks must co-evolve with LLMs to guide their progress in a direction that benefits humanity.

There are still many open questions and challenges around evaluating and applying LLMs responsibly. But continued research, open and constructive dialogue, and proactive efforts can help unlock immense possibilities while mitigating risks. The extent to which we realize the benefits of LLMs versus their pitfalls is ultimately tied to the values and priorities of those building, governing and using them. A mindset of responsibility, care and wisdom will be especially important.

Finally, we would like to share a summary of recommendations that we have already made in various places of this document, which might help in considering how to implement LLM-based solutions for addressing LMIC needs.

11. Recommendations for LLMs in LMICs

Our primary recommendations can be categorized into three parts: localization, evaluation, and deployment.

11.1. LOCALIZATION

Localization and Contextualization

- We propose the curation of a set of local documents in different geographies (State level not just National level) and in different domains (Health, Agriculture, etc.) that

embody current best practices or specific diseases / conditions that are known to have a high prevalence or other determination of importance with respect to impact. Global guideline documents (e.g., WHO ANC guidelines) can be a starting point, but local stakeholders from CHWs to MoH advisors typically have their own guidelines borne out of pragmatism to optimize impact in their own context. LLM solution providers can then use these curated data sets for their own use cases.

- Curated documents should be LLM-ready. For example, crucial information in WHO's ANC guidelines often appear in tables in documents in picture format. Text-based LLM solutions will not ingest this crucial information.

Local language and Vernacular Communication

- We propose that LLMs are designed to be optimized for local language and for vernacular (everyday language, such as Nigerian Pidgin). As a starting point, it will be useful to curate a reference manual on what languages and vernacular communication are used by different high-priority groups (e.g., pregnant women in Nigeria, or farm workers in Tamil Nadu, India) so that LLM-based solution developers can retrieve and use the information.
- We also propose the curation of another reference manual on documenting the cultural aspects of which digital communication methods (e.g., real-time voice over phone, SMS, voice recording, WhatsApp) are preferred ways of communicating, categorized by at least region and priority group type (e.g., pregnant women, farm workers).

11.2. DEPLOYMENT

Solution Integrity

- We recommend not to assume that a technology developer in isolation understands the technical requirements for LLMs even in their own or other LMIC region.
- We recommend coordinated oversight and development of baseline digital infrastructure that supports LLMs in all LMICs with requirements input from stakeholders throughout all LMICs of interest. We recommend identifying and designing baseline digital infrastructure to support LLMs for the worst case while not limiting performance in the best case.

Scalability

- We recommend leveraging local software talent in individual LMICs to configure and maintain software.
- As well as leveraging local talent, we recommend leveraging software talent worldwide (from both LMICs and non-LIMCs) so that both reliable and world-class and newly developed core software capabilities are made available to individual LMICs, and,

crucially, designed in such a way that the software is reusable so that a wide range of talent in LMICs can build upon them.

- We propose that baseline digital infrastructure for LLMs may include voice or text data capture and transmission (either offline then online, or intermittent online), LLM server response (either offline then online, or intermittent online), data sharing (voice or otherwise) across different LLM use cases, as well as higher-level functions such as language translation.
- We suggest that the same development methods used to develop open source repositories such as Open Health Stack[51][53], as well as the open-source code base itself, can be a starting point to add core software infrastructure tools for LLMs for LMICs.

Adoption

- We recommend considering change management between old processes and new processes that involve LLM-based tools as a programmatic effort by itself, to achieve sustained adoption.
- We recommend developing technical capabilities and workflow processes such that old and new processes can still work together while problems with new capabilities are resolved, and trust is gained.
- We recommend ensuring that every stakeholder across the spectrum (e.g., patients to HCWs to senior Government officials) realize an overall benefit by the addition of the new capability.

Sustainability

- We propose identifying, developing or adopting digital standards in public solutions that are also used in private solutions that are designed for customers at a higher end of the socio-economic scale in a region.
- We recommend developing technologies that are reusable and low-cost to maintain and support since it means that it will minimize the private economic activity or donor funding that will need to occur to subsidize or pay for the public infrastructure.
- We recommend developing digital infrastructure for LLMs that can work across multiple use cases and allows for data leverage, enabling powerful LLM use cases that can take into account personal information across multiple siloed data repositories.

11.3. EVALUATION

LMIC specific metrics for LLM capabilities

- We propose the curation of specific LMIC-specific metrics for LLM capabilities. For example, in the case of giving pregnant women in Nigeria an equal opportunity to go

to a clinic, an LLM metric may be based around: a) the receiver operating characteristic (ROC) curve of the ability of the LLM to identify from an inbound or outbound health query that the woman is pregnant, and b) the ROC curve of the ability of the conversational language of the LLM to convince the woman to come to the clinic during her pregnancy. Outcome metrics are different to these metrics, although it is expected that positive LLM metrics will feed through to positive outcome metrics. Long-term analysis of outcomes can be used to evaluate this latter hypothesis.

An overall program for evaluation of LLM solutions

Broader than specific metrics for LLM capabilities in LMICs, we propose an overall program for the evaluation of LLM solutions. Such a program may include metrics that quantify potential barriers to the adoption and scaled impact of an LLM solution, many of which have been discussed in this document. For example, the LLM-based solution may provide an answer that is technically optimal, but if an FLW is not authorized to perform that function, then the LLM solution is suboptimal.

APPENDIX A - THE LEARNING MECHANISMS OF LLMS

The learning mechanism of Large Language Models (LLMs) is a fascinating aspect of their design, yet it remains an open research question. Unlike logic-based systems, where the developer can introspect and narrate how the system achieves a task, LLMs learn the rules directly from demonstrations or examples. The developers do not encode any rules inside the system, which makes understanding the internal workings of these models a complex endeavor.

How Do LLMs Learn?

LLMs learn by being exposed to vast amounts of text data, including textbooks, internet forum discussions, news articles, social media posts, and fiction. They are trained to predict the next word or sentence given a piece of text, a process that requires them to understand the context and nuances of the language.

An analogy to illustrate this process is a mystery novel; if the model is shown all the evidence right before the mystery is revealed and is asked to solve the mystery, it would need to understand many aspects of the story to achieve this task. When a training process is scaled up to train on millions of mystery stories where the conclusions are known, the idea is that a certain depth of apparent understanding begins to emerge within the system such that the LLM would be able to solve a new mystery given the evidence.

The Challenges of Understanding LLMs

Despite their impressive capabilities, LLMs present significant challenges in terms of understanding how they work. These models are opaque mathematical objects, which makes it difficult to certify and ensure reliable performance.

Moreover, LLMs are not grounded in physical reality, meaning they do not distinguish between facts and fiction. They treat all text completion tasks equally, whether they are based on facts or fictional stories. This can lead to the propagation of misinformation if not properly managed.

The Limitations of LLMs

Despite their impressive learning capabilities, LLMs have their limitations. They can sometimes produce harmful or hateful text, which is why the fine-tuning stage of training is crucial to mitigate such risks.

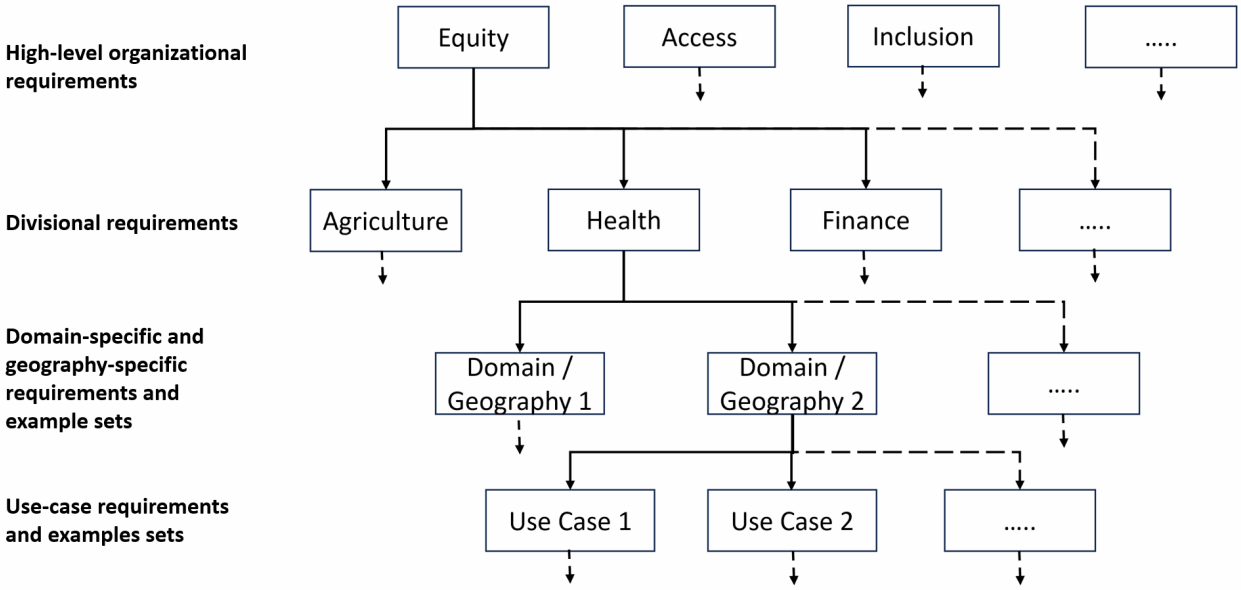
Furthermore, without safety guardrails, an LLM base model will attempt to answer any question, even those that could potentially be harmful or illegal.

In conclusion, while LLMs have a remarkable ability to learn from text data and generate meaningful outputs, understanding their learning mechanism remains a challenge. Their limitations also highlight the importance of careful training and evaluation to ensure their safe and effective use.

APPENDIX B - A HIERARCHICAL REQUIREMENT AND EVALUATION FRAMEWORK

One of the challenges in developing metrics and evaluating LLMs on those metrics is that the definitions of the metrics may vary according to domain, geographical location, cultural factors, and use-case. Hence, there is a risk that the metrics may be inconsistent, and the problem of evaluation may become intractable. In particular, the benchmarks used for evaluation across the different domains may not address all the concerns embedded in the metrics.

One possible way to maintain consistency in LLM metrics and the evaluation process across highly diverse use cases in different geographies is to consider a hierarchical metrics and evaluation framework.



At the top of the hierarchy are values and requirements that an organization may require at a mission-statement level. These can provide a set of base categories for defining and using the metrics. At this level, we can identify the core issues that are common and universal across domains and locations, and define a base set of metrics according to those. These can potentially help in developing a common set of benchmarks as well.

The next level in the hierarchy addresses domain-specific and geography-specific priorities and example sets. This is designed to target requirements and evaluation criteria focused on topic areas that local MoHs and NGOs are usually aware of as being contentious. For example, in healthcare in certain countries, ward keepers and clinic workers each play a role in the domain of ANC care but there can be tension between the groups due to differences in how they are recognized and managed by the local MoH. These priorities and the examples that motivate them need to be in a form that

can be ingested technically into solutions developed in multiple use cases in the next layer of the framework hierarchy. For these different use cases, appropriate benchmarks would have to be established.

The final level in the hierarchy addresses use-case-specific priorities and example sets. For example, consider a “family-planning” use case in an “ANC / Nigeria” domain under a “health” and “equity” evaluation criteria. This use-case example is quite complicated since cultural dynamics mean that husbands can refuse to use birth control methods when asked by their wives, and husbands can potentially put pressure on their wives not to use birth control, against the women’s wishes. However, local MoHs and many NGOs have dealt with these dynamics for many years and have often developed strategies and workflows that manage such situations. Assuming the capability of an LLM component at least matches the capability of existing accepted manual or machine-guided strategies and workflows, then the metrics of success for “equity” (in this example) can include the desired strategy or workflow result that is currently accepted and in use today.

One benefit of such a hierarchical requirement framework is that the previous layer in the hierarchy can be used as a starting point for defining sub-requirements, which saves effort since requirements can be re-used rather than re-invented for each use case and ensures that there is some consistency in requirements across different use cases.

APPENDIX C – RECOMMENDATIONS AND PROPOSED NEXT STEPS

Summary of Recommendations

Below is a table describing the core recommendations for LLMs in LMICs given in Section 11, separated into three parts: Localization, Deployment, Evaluation.

Localization	Deployment	Evaluation
• Incorporate local and contextual information. • Enable communication in local language, dialect, and vernacular. • Ensure oral communication	• Adapt to constraints in digital infrastructure and data pipelines. • Incorporate change management programs. • Foster local SW developer ecosystem with technical oversight from global SW groups	• Develop LLM metrics specific to use cases in LLMs. • Develop metrics for Diversity, Equity, and Inclusion • Develop evaluation methods against systemic constraints on roles of various actors

Suggestions for how to implement the recommendations

LOCALIZATION

Contextualized knowledge

- We propose the curation of a set of local documents in different geographies (State level not just National level) and in different domains (Health, Agriculture, etc.) that embody current best practices or specific diseases / conditions that are known to have a high prevalence or other determination of importance with respect to impact. Global guideline documents (e.g., WHO ANC guidelines) can be a starting point, but local stakeholders from CHWs to MoH advisors typically have their own guidelines borne out of pragmatism to optimize impact in their own context. LLM solution providers can then use these curated data sets for their own use cases.
- Curated documents should be LLM-ready. For example, crucial information in WHO's ANC guidelines often appear in tables in documents in picture format. Text-based LLM solutions will not ingest this crucial information.

Local language and Vernacular Communication

- We propose that LLMs are designed to be optimized for local language and for vernacular (everyday language, such as Nigerian Pidgin). As a starting point, it will be useful to curate a reference manual on what languages and vernacular communication are used by different high-priority groups (e.g., pregnant women in

Nigeria, or farm workers in Tamil Nadu, India) so that LLM-based solution developers can retrieve and use the information.

- We also propose the curation of another reference manual on documenting the cultural aspects of which digital communication methods (e.g., real-time voice over phone, SMS, voice recording, WhatsApp) are preferred ways of communicating, categorized by at least region and priority group type (e.g., pregnant women, farm workers).

Develop a self-sustaining Content Management System (CMS) as a Digital Public Good (DPG) for domain and locality specific content.

A self-sustaining, domain-specific Content Management System (CMS) for maintaining and curating local content can be developed. Below are some ways that this can be used in the Healthcare domain. The processes and technical solution for the developed CMS (with different data) can be used for AgTech, Financial services for the poor, and education sectors.

As an example, in Healthcare, the CMS can be used:

1. to maintain videos, documents, images, including but not limited to applicable WHO and MoH guidelines.
2. to tag content based on sector, country, state, region, area of concern etc.
3. as a platform for in-country translation of any non-English content which is used as guidelines in country/state/region
4. by MoH users to version/revise the content
5. to maintain transcriptions for video contents as well language translations.
6. as a Digital Public Good allowing various MoHs to both manage their content as well leverage others
7. so that NGO's can contribute to it.
8. to generate local employment opportunities in curating / translating / collecting / developing content
9. to provide cross LMIC inspiration for organized content
10. by LLMs to develop embeddings databases for localizing their responses.

Optionally, local Govts who may control the CMS can also investigate leveraging curated content to allow private operators to get access to reliable and curated content, potentially for a sustainability fee, and get feedback from various public usages of the system.

LLM-based solutions would be a client to this system and would use contextual (country, state, region, area of concern) queries to augment the LLM responses.

Eventually the same CMS can also be used to fine-tune the LLM's models.

As a very first step, 3-5 core domains and use cases (e.g., within health and agriculture) can be used as focused streams of activity around which the processes to develop the CMS can be built up.

Develop SW Tools for use by LLM-based solutions that bridge the gap between vernacular language and English (both spoken and written).

Commercial companies are already trying to develop best of breed models for text to speech, speech to text and translation(text and speech) models. Given the scale and magnitude of the development task, we suggest that organizations leverage these efforts and collaborate with the companies to try and ensure that languages important to LMICs are addressed substantially. In absence of interest and sufficient capabilities from companies, specialized speech conversion models can be developed for each region.

Local, privacy-sensitive, consented campaigns can be started to collect domain specific datasets in LMICs e.g., patient to HCW conversations in vernacular language.

Deployment

Solution Integrity

- We recommend coordinated oversight and development of baseline digital infrastructure that supports LLMs in all LMICs with requirements input from stakeholders throughout all LMICs of interest. We recommend identifying and designing baseline digital infrastructure to support LLMs for the worst case while not limiting performance in the best case.

Deploy in-country instances of LLM inference engines.

A key aspect of adoption is availability and access. Today's foundational LLMs models are far too expensive to run by small groups/Universities even just for inference e.g., Falcon180B model requires approximately US$42/hr at the base minimum. Multiple concerns exist in using hosted services like OpenAI/HuggingFace/etc. , for example, fear of losing citizen-private data to commercial companies, as well as hourly usage costs.

Efforts can be bootstrapped to create in-country instances of these models. These LLM instances do not have to be highly scalable initially but are created with the idea of provisioning so that a large community of local developers and academics can use the instance for an LMIC appropriate fee. An example of a very low cost LLM instance from TextSynth is here: TextSynth pricing. In this case costs are on a per-token basis and not an hourly basis.

Catalytic programs in each region could support:

1. Initial setup efforts
2. Initial marketing costs to bring enough usage to sustain running costs
3. Cost of running the Digital Public Infrastructure (DPI) until a breakeven point.

It is proposed that this should not necessarily be a free Digital Public Good but a pay per use service, even if the cost is minimal per use, since in aggregate there is then a potential sustainability model.

Build an ecosystem of tools for LLM usage for developers in LMICs

Regardless of which LLM strategy is chosen, a basic digital framework needs to be in place to send, store, process and receive data. For example, there are emerging software standards such as WHO's SMART Guidelines in the healthcare domain that depend on software infrastructure that enables interoperability between different systems, as well as managing data transfers in offline, online and intermittent use cases, which are prevalent in LMICs. It is out of scope of this paper to review these, but such existing capabilities in each domain should be leveraged and built upon where possible to create an ecosystem of reusable tools for developers in LMICs to utilize.

Scalability

- We recommend leveraging local software talent in individual LMICs to configure and maintain software.
- As well as leveraging local talent, we recommend leveraging software talent worldwide (from both LMICs and non-LIMCs) so that both reliable and world-class and newly developed core software capabilities are made available to individual LMICs, and, crucially, designed in such a way that the software is reusable so that a wide range of talent in LMICs can build upon them.
- We propose that baseline digital infrastructure for LLMs may include voice or text data capture and transmission (either offline then online, or intermittent online), LLM server response (either offline then online, or intermittent online), data sharing (voice or otherwise) across different LLM use cases, as well as higher-level functions such as language translation.
- We recommend that the same development methods used to develop Open Health Stack[51][53], as well as the open-source code base itself, can be a starting point to add core software infrastructure tools for LLMs for LMICs.

Develop an LLM-for-LMIC developer network.

This is to set up processes for training SW developers in LMICs on how to use and build up LLM tools. There would be a train-the-trainer component, creating scalability of AI-trained resources. Having a core curriculum based around common and curated tools, such as the proposed Open LLM Stack (OLS), will make adoption, scalability, and sustainability of such a developer network possible.

Developers will be able to start with components that are known to be low-cost, functional, and scalable. This can be modeled on the existing and quite successful Open Health Stack developer model.

Adoption

- We recommend considering change management between old processes and new processes that involve LLM-based tools as a programmatic effort by itself, to achieve sustained adoption.
- We recommend developing technical capabilities and workflow processes such that old and new processes can still work together while problems with the new technologies are resolved, and trust is gained.
- We recommend ensuring that every stakeholder from patients to HCWs to senior Government officials realize an overall benefit by the addition of the new capability.

Include Change Management to manage adoption between existing and LLM-based capabilities as part of all programs.

All LLM-based programs should have a component to show how they will transition from the existing process flows to the new process flows that incorporate LLMs.

The dynamics of what the responsibilities of each stakeholder are both before and after the LLM intervention need to be defined, as well as how the intervention allows the stakeholder to maintain their responsibilities without risk during adoption.

Some specialized technologies may need to be developed especially to manage the transition from the old to the LLM-based intervention.

Sustainability

- We propose identifying, developing, or adopting digital standards in public solutions that are also used in private solutions that are designed for customers at a higher end of the socio-economic scale in a region.
- We recommend developing technologies that are reusable and low-cost to maintain and support since that minimizes the private economic activity or donor funding that needs to occur to subsidize or pay for the public infrastructure.
- We recommend developing digital infrastructure for LLMs that can work across multiple use cases and allows for data leverage enabling powerful LLM use cases that take into account personal information across multiple siloed data repositories.

Seek private and commercial use cases for Open LLM Stack and link them to public use cases.

The ability to maintain a linkage and leverage between commercial activity in LLMs in LMICs as well as public-sector activity in LLMS in LMICs we believe is very important from several perspectives. Firstly, from the viewpoint of regional uplift and training talent on LLMs, having similar standards and code bases in both the public and private sectors in the LLM development space means that workers can freely move between the public and private sector, providing a synergistic ecosystem. Second, in terms of financial sustainability, having similar LLM-based

components in private use cases makes it easier for them to be used at a much lower cost or completely subsidized in public use cases.

Evaluation

Broader than specific metrics for LLM capabilities in LMICs, we propose an overall program for the evaluation of LLM solutions. Such a program may include metrics that quantify potential barriers to the adoption and scaled impact of an LLM solution, many of which have been discussed in this document. They should address the key aspects of Equity, Access and Inclusion as well as the eventual healthcare impact of the solutions. For example, in the case of giving pregnant women in Nigeria an equal opportunity to go to a clinic, an LLM metric may be based around a) the ROC curve of the ability of the LLM to identify from an inbound or outbound health query that the woman is pregnant, and b) the ROC curve of the ability of the conversational language of the LLM to convince the woman to come to the clinic during her pregnancy. Outcome metrics will be specific to the use-case and solution, although it is expected that the EAI metrics will feed through to positive outcome metrics. Long-term evaluations of outcome metrics will be able to validate this hypothesis.

Set up a program for the overall evaluation of LLM solutions.

This can begin by the curation and ranking of potential LLM use cases, together with a curation and ranking of currently available LLM capabilities and evaluation criteria (some of which has been performed in the landscape document).

The proposed next step is to then identify a set of 3-5 core LLM use cases that can be used to jumpstart developing a program for the evaluation of LLM solutions. So, concurrent with work being performed on the actual LLM use cases (using Open LLM Stack and the other reusable capabilities proposed earlier), a set of evaluation processes would be developed in parallel, using the 3-5 programs.

The 3-5 programs should be geographically diverse, as well as cover different verticals (healthcare, agriculture) to be representative of the domains that should be covered by the evaluation processes.

Summary of Implementation Steps

Below is a summary of the implementation steps proposed above.

Localization	Deployment	Evaluation
• Develop a self-sustaining Content Management System (CMS) as a Digital	• Deploy in-country instances of LLM inference engines.	• Set up a program for the overall evaluation

Public Good (DPG) for domain and locality specific content. • Develop SW tools for use by LLM-based solutions that bridge the gap between vernacular language and English (both spoken and written).	• Build an Open LLM Stack (OLS) to create an ecosystem of tools for LLM usage for developers in LMICs. • Develop an LLM-for-LMIC developer network. • Include Change Management to manage adoption between existing and LLM-based capabilities as part of all programs. • Seek private commercial use cases for Open LLM Stack and link them to public use cases.	of LLM-based solutions.

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